

STATISTICS FOR MICROFOSSIL CONCENTRATION MEASUREMENTS EMPLOYING SAMPLES SPIKED WITH MARKER GRAINS

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ABSTRACT

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Microfossil abundance in sediment is often expressed as the proportion each taxon comprises in the total assemblage. A different way to calculate abundance involves adding a known quantity of exotic markers to a measured quantity of sediment and using the markers as a standard to which the indigenous taxa are compared. By this means a taxon's concentration (microfossils per cm^3 or per g) and influx (microfossils per cm^2 per yr) can be estimated. Although such measurements have potential value, they are all but useless unless experimental error can be assessed.

This paper uses examples from palynology to describe a method using parametric statistics for assigning confidence limits to the true concentration of a taxon in a single sample of sediment. I discuss options for keeping the confidence intervals as small as possible. A test of the null hypothesis that the concentration of a taxon is the same in two independent samples is presented, and a method is developed for assigning confidence limits to the true concentration of a taxon in N independent samples which are combined. The exotic marker technique produces a positive null model correlation coefficient for the concentrations of the separate taxa; an observed correlation coefficient should be tested to see whether it differs significantly from the nonzero constraint coefficient. The concept of "influx index" is introduced to acknowledge the ambiguities caused in influx data by uncertainties in sedimentation rates.

INTRODUCTION

Shepherds often estimate the size of their herds by counting only the black sheep and multiplying that number by the ratio of white sheep to black that exists in their flock. In a similar manner, rough wildlife censuses are carried out by tagging a number of individuals and then releasing them into the unmarked population. If the tagged members distribute themselves throughout the population, the size of the population can be inferred by sampling its members and recording the ratio of untagged to tagged individuals in the samples.

Measurements are carried out in many fields of science by adding a known amount of a recognizable substance to the sample and calculating the

abundance of material indigenous to the sample as a multiple of the standard. A sample containing a standard is often said to be spiked; a “spike” is a colloquial term for alcohol added to a soft drink.

The spike technique is useful in certain of the Quaternary disciplines. Maher (1972, pp.537–538) summarizes its development by palynologists who employ exotic marker grains to estimate pollen concentration, the number of pollen grains in a unit volume or weight of sediment. If sedimentation rates are known, concentration estimates can be converted to influx, a measure of the number of pollen grains incorporated into the sediment per cm^2 per yr. The technique can be adapted to estimate the concentration of other things, such as diatoms, other algae, fungi, larvae, charcoal fragments, volcanic ash shards, mineral species, phytoliths, etc. Considering the potential of the method, it is surprising it is not more widely used.

I will discuss certain of the problems and possibilities of the spike technique in this paper, using examples from palynology. Although for the most part I will consider pollen concentration in $\text{grains}/\text{cm}^3$ of wet sediment, those interested in the concentration of objects/gram of sediment should simply change the sediment measure from cm^3 to grams.

The concentration of fossil pollen in a sample is generally based on the proportion:

$$\frac{\text{Fossil pollen in sample}}{\text{Fossil pollen counted}} = \frac{\text{marker grains in sample}}{\text{marker grains counted}}$$

Let M refer to the number of marker grains added to a vial containing the sediment sample, and let x be the number of grains of a taxon X counted while tallying n marker grains in a slide made from the material in the vial. Taxon X can be any pollen taxon in the sample, any groups of taxa, or the sum of all types. Finally let $R = x/n$, the ratio of pollen to markers in the count. We see that the number of grains of taxon X in the sample $= RM$. If V is the volume of the sediment sample, the concentration of taxon X in $\text{grains}/\text{cm}^3 = RM/V$. And if S is the sedimentation rate in cm/yr for the interval from which the sample was taken, then the influx of taxon X in $\text{grains}/\text{cm}^2/\text{yr} = SRM/V$.

The concepts of pollen concentration and pollen influx are attractive to pollen analysts because the trend of a single pollen taxon through time can be considered in a frame of reference that is independent of the other members of the pollen assemblage. The number of oak pollen grains/ cm^3 may increase if there are more oak trees in the area or if they are closer to the site of deposition. Oak concentration might also increase if the sedimentation rate slowed, so that more pollination years were included in the sample. The increase might also be related to a change in the type of sediment or a change in the sedimentary environment such that fewer grains were destroyed after they were buried. There could be other reasons as well. If the influx of oak pollen increased, the analyst could rule out the effect of a changing sedimentation rate, but the other possible causes would remain for consideration. Pollen concentration and influx will not answer all our questions, but they

can help to view the problem from perspectives not possible with percentages. The best prospect for solving a problem lies in looking at the data from as many perspectives as possible.

The terms $M/(nV)$ and $SM/(nV)$ will be constant for all taxa in an individual sample. If these constants are multiplied by the number of grains of any taxon X in the count from that level, the product will equal, respectively, the concentration and influx estimates for that taxon. Such estimates can be obtained rapidly with the simplest calculator.

But how good are these estimates likely to be? Analysts who have never studied replicate samples from the same level in a core tend to fall into a trap; they attach significance to any fluctuation that appears in their data. Those who know that concentration data are inherently "noisy" often err in the opposite direction, ignoring even significant changes. In fact, it is not possible to know whether a taxon's concentration has changed through time, unless we are able first to judge how much uncertainty is involved in the estimate itself. The problem is rather like trying to judge the effects of a diet by weighing oneself each day on different, uncalibrated scales.

So far we have spoken as if R , M , V , and S were known sample constants. This is obviously not true. The number of grains M with which the sample is spiked can never be known with certainty, nor can R , V , and S . I have in fact used these symbols to emphasize that they are actually variables and not constants. We must deal with the three variables R , M , and V in order to measure pollen concentration. Later, when I discuss pollen influx, we will need also to consider S .

Several workers (see Regal and Cushing, 1979, and references cited therein) have suggested methods for assigning confidence intervals to concentration estimates. Some require slavish adherence to fixed-sum counts to avoid constant reference to statistical tables. Others require special counting procedures that differ from those normally used in measuring pollen proportions.

The analyst needs to be able to make his counts in the standard manner, and at the end of the counting period he should have all the information to present the data either as proportions or as concentrations or as influx. I will describe a method of calculating a confidence interval for the true concentration of a taxon at any level in a core given a single sample from that level. I discuss implications of the positive null model correlation coefficient of the separate taxa's concentrations that is imposed by the spike technique. I provide a test to judge whether an observed change in concentration between levels is significant. I develop a method for assigning confidence limits to the true concentration of a taxon in N independent samples which are combined, and discuss strategies for keeping confidence intervals as small as possible. Finally, I will make some comments about influx measurements and introduce the concept of an "influx index".

Statistical methods can define the inherent uncertainty in accurate work; they cannot improve faulty data. Therefore in the following discussion I will always assume that the pollen is well preserved and correctly identified. I will also stipulate that the analyst has flawless technique and produces slides in

which the assumptions of Bernoulli trials are met. Finally I will assume that when the counts for a taxon X are inside the pollen sum they will follow a binomial distribution; when the counts are outside the sum they will follow a negative binomial distribution (Mosimann, 1965, p.637).

CHOOSING THE SIZE OF THE SPIKE

In the normal practice of estimating pollen concentration by the addition of a marker grain standard, the pollen analyst will spike the sample, process it, and then count to a set number of indigenous pollen grains while also tallying the marker grains. The size of the indigenous pollen count x will reflect the analyst's opinion as to how large the pollen sum must be to yield reasonable estimates of the proportions of the constituent taxa. (The number of indigenous pollen for the ratio count will become the pollen sum when the standard fossil pollen proportions are calculated.)

When the measured sediment sample is to be spiked with exotic marker grains, the following question naturally arises: How many marker grains M should be added to achieve the most precision for the least work? For purposes of this discussion, I will consider "work" to be directly proportional to the total number of indigenous pollen x plus marker grains n that are counted. "Precision" involves the degree of certainty to which the true ratio of pollen to markers in the sample can be inferred from a count. Mosimann (1965, p.659) calls the true ratio u , a common symbol for a ratio, and he provides a method to set 0.95 confidence limits on u based on $\hat{u}$, the mean ratio of the number of indigenous grains to marker grains that were counted in the sample: $\hat{u} = x/n = \bar{R}$. The most precision occurs when the ratio's confidence interval is small relative to the value of the ratio itself. That is to say, precision improves as the quantity "length of the confidence interval/ $\hat{u}$ " decreases.

The analyst can control the indigenous/marker grain ratio by the quantity of marker grains introduced to the sample, so the problem involves choosing the best ratio. Simple intuition suggests that a ratio will be most informative when the numbers in the numerator and denominator are simultaneously as large as possible; of course, this implies that the marker grains should be as abundant as the indigenous pollen in order that $u = 1$ (Regal and Cushing, 1979). I have used Mosimann's equation for outside counts to calculate 0.95 confidence limits for u based on assumed values of x and n . For any specified level of work (that is, for any value of $x + n$), the 0.95 confidence interval/ $\hat{u}$ achieves a minimum value when $\hat{u} = 1$. This result is not very good news for the practicing pollen analyst because it means that only half of the grains that are counted will provide any information about the fossil community; the other half will be exotic marker grains. Although the marker grains are necessary for measuring concentration, they do not help to resolve the proportions of the indigenous taxa.

I will now show that the ratio that provides the most precision for a specified amount of work is not necessarily that which will maximize precision and information while minimizing counting effort. Fig.1A illustrates a case where

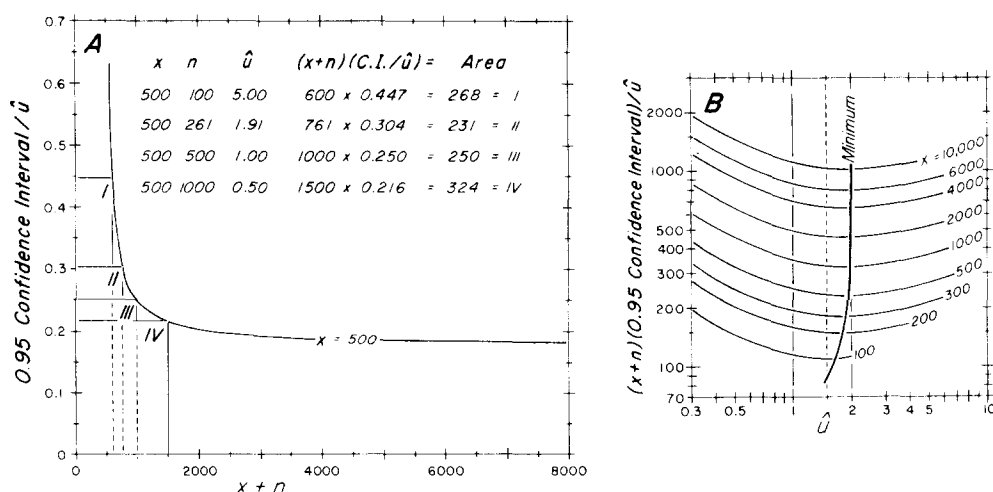


Fig.1. A. The effect of counting work on the relative precision of a ratio (x/n) when $x = 500$. B. The most efficient indigenous pollen/marker ratio ($\hat{u}$) to achieve the most precision for the least work ≈ 2 .

the analyst plans to count 500 indigenous grains per sample in order to use this as the pollen sum upon which pollen proportions (inside counts) will be based. The analyst wishes to decide how many markers to add to measure pollen concentration. The curve shows how the precision (0.95 confidence interval/ $\hat{u}$) varies with counting work ($x + n$). Because $x = 500$, this also sets a lower limit to $x + n$. (Mosimann's equation was not used to calculate ratio limits when $n < 50$; the curve of Fig.1A extends over the range of $x + n$ from 550 to 8000.) The curve can be thought of as the locus of points formed by the upper right corners of some 7450 rectangles with various dimensions of $[0.95 \text{ confidence interval}/\hat{u}]$ and $[x + n]$, four of which are shown in Fig.1A. The rectangles differ in area ($[x + n] \times [0.95 \text{ c.i.}/\hat{u}]$), and the one with the minimum area yields the highest precision for the least work. This rectangle is labeled II. Rectangle I offers lower counting effort, but lacks precision, and rectangles III and IV provide good precision but require too much work. I mentioned previously that the ratio $\hat{u} = 1$ will provide the best precision for a set amount of work; if $x = 500$, then a total of 1000 pollen and marker grains must be counted (rectangle III). Fig.1A shows that for $x = 500$, the most efficient $\hat{u} = 1.91$ (rectangle II); good precision can be obtained with about half the number of marker grains that are required when $\hat{u} = 1$. Fig.1B is a plot of $[x + n] \times [0.95 \text{ c.i.}/\hat{u}]$ versus the indigenous pollen/marker grain ratio $\hat{u}$ for values of x from 100 to 10,000. As x increases, the curves of "area" obtain their minimum values as $\hat{u}$ approaches 2.

When a sample's $\hat{u} = 2$, two-thirds of the grains in the count will provide information about the proportion of the taxa in the fossil assemblage. By adjusting $\hat{u}$ to 2 rather than 1, the analyst simultaneously maximizes information about the fossil assemblage, maintains a good level of precision for concentration estimates, and reduces the total counting effort.

It is obviously not possible to maintain an optimum value of u in each sample in a sediment core because one would have to know in advance the very information that the study was designed to yield. However, the analyst is well advised to experiment with a few replicate samples from different levels in the core. An arbitrary number of marker grains can be added to the measured sample, and a rapid pollen count can then establish a rough estimate of u . With this information, an informed decision can be made as to the number of markers that will be required to yield a pollen/marker ratio of about 2. If the pollen concentration differs at different depths in the core, it may be advisable to vary the marker grain spike in the several zones. The point is that it makes little sense to take a sample, add a spike, and then hope that there will be about the right ratio of indigenous to marker grains on the slide. The more the ratio differs from 2, the larger will be the penalty the analyst pays; and that penalty will be measured in the added days and weeks spent at the microscope.

MARKER GRAINS IN TABLET FORM

In the preceding section, I have shown that estimates of pollen concentration are most efficiently obtained when the sediment sample has an optimum indigenous pollen/marker grain ratio. The analyst can control this ratio by changing the size of the measured sediment sample and by varying the number of marker grains with which the sample is spiked. The marker grains are often suspended in a liquid, and the average concentration of markers per volume is measured. A certain volume of this standard mixture is then pipetted into each sediment sample. The number of marker grains that were added is considered to be proportional to the volume of the spike.

Stockmarr (1971) introduced a different and ingenious technique by mixing acetolyzed *Lycopodium* spores with calcium carbonate and lactose and pressing the powder into tablets. He later produced a large supply of tablets (batch no. 212761), and used a Coulter Counter to estimate the spore content per tablet. In the 71 tablets he counted, the mean number of spores per tablet $\bar{Y}$ was 12,489 with a standard deviation s of 491 (Stockmarr, 1973, values calculated from data in fig. 2, p.89). The tablets offer a much more convenient method of introducing marker grains to a sample than does the older pipette technique. But a question arises when tablets are used. If several, say N , such *Lycopodium* tablets are added to a pollen sample, what is the expected total number of spores in the spike, and what is the best estimate of this number's standard deviation? The solution is not difficult, but it has very important implications for measuring concentration.

If repeated random samples of N *Lycopodium* tablets are drawn from the parent population that has mean $\bar{Y}$ and standard deviation s , the frequency distribution of the sample means $\bar{Y}_N$ in these repeated samples has mean $\bar{Y}$ and standard deviation $s/\sqrt{N}$ (Snedecor and Cochran, 1967, p.50). For a single group of N tablets, then, $\bar{Y}$ is an unbiased estimate of the expected sample mean, and the expected standard deviation of the sample mean, or

the standard error of the mean, is $s/\sqrt{N}$. The expected number of spores in a combined group of N tablets is $N\bar{Y}$ with expected standard deviation $Ns/\sqrt{N}$. That is, the expected standard error of $N\bar{Y}$ is equal to $Ns/\sqrt{N} = \sqrt{N} \cdot s$. The variance of the spore number in a group of N tablets = Ns^2 , which, as we might have guessed, is simply the sum of the variance contributed by the individual tablets in the group.

Stockmarr's *Lycopodium* tablets offer a simple method of increasing the precision with which the marker grains in a sample can be estimated. If a single tablet is added, the spike mean is 12,489 with standard deviation 491, and the coefficient of variation $100s/\bar{Y} = 3.9\%$. If four tablets are added, the spike mean is 49,956 with standard deviation 982, and the group coefficient of variation drops to 2%.

VOLUME SAMPLES

For practical reasons, pollen samples generally are smaller than 2 cm^3 . If "pollen grains/ cm^3 " is to have any real meaning, concentration measurements must be carried out in sediment that is homogeneous at the scale of the samples. This generally restricts such studies to water-saturated sediments with particles of silt and clay size. Sediments may differ markedly, and a variety of devices and techniques have been used to take volumetric samples; these range from calibrated spoons, tubes, boxes, capsules, and syringes to water-displacement methods.

I once concluded (Engstrom and Maher, 1972, p.356) that small cylindrical samples have a relatively large surface to volume ratio, and that surface irregularities would limit the reproducibility of very small samples. In light of the foregoing discussion of marker tablets, it should also be possible to improve sediment volume measurements by combining several small sample aliquots. I have found that disposable 1 cm^3 tuberculin syringes make handy, cheap devices to sample plastic clays.

Cut off the tips from several dozen syringes, and make a single metal spacer like that shown in Fig.2C. The spacer, the only precision part, is used to control the plunger movement that expells the sample aliquot. Modify a couple of 3 cm^3 plastic syringes (Fig.2A, B) to load the samplers. Syringe A is used as a miniature piston sampler to take sediment from a core. The plunger is removed from syringe B, and the sediment is injected into its base. Syringe B is then used to inject small amounts of sediment into the bases of several of the tuberculin syringes. Examine the sediment as it is pushed through the length of the small syringes, and discard any with air spaces. The technique takes longer to describe than to do. Given enough syringes, it is possible to sample a number of levels before stopping to wash the equipment. The rubber seals can be pulled off the plungers for cleaning. After the last use of the day the rubber should be lubricated. Dip the plungers into a mixture of tertiary butyl alcohol and silicone fluid (~ 1 drop silicone per ml of TBA); the alcohol evaporates leaving the rubber coated with a thin lubricant film.

I calibrated my sampler by taking replicate samples of modeling clay of

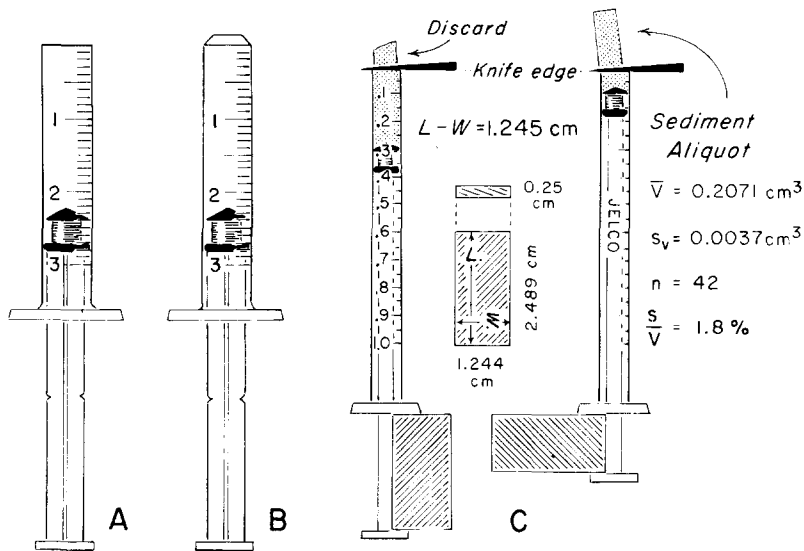


Fig. 2. Disposable plastic syringes as sediment samplers. A. Small piston sampler made by cutting off the end of a 3 cm³ syringe. B. Sediment injector made by partially cutting off the tip of a 3 cm³ syringe. C. Volumetric sampler made from 1 cm³ tuberculin syringe with end removed. Rectangular metal spacer controls plunger movement.

measured specific gravity. The single aliquots approximated a normal distribution with mean 0.2071 cm³ and standard deviation 0.0037 cm³. The coefficient of variation = 1.8%. If samples were made combining four such aliquots, they would have mean 0.8284 cm³ and standard deviation 0.0074 cm³; the coefficient of variation would drop to 0.9%.

It is good practice to take several aliquots when sampling a core even if the analyst does not plan to combine them into a single sample. One could be used to choose a proper spike; another could be weighed wet and dry for an estimate of water content. In fact, if the aliquots were all weighed the analyst could calculate pollen concentration both in grains/cm³ and grains/g. All sediment aliquots should be taken when the core is fresh; the time spent in this activity is but a tiny fraction of that spent to process and count the samples.

MARKER GRAINS/VOLUME: THE VARIABLE M/V

When concentration measurements are based on the spike technique, the abundance of the indigenous grains in the sediment is taken as some multiple of the sample's artificial concentration of marker grains/cm³; the quantity M/V is generally assumed to be constant. Let us imagine that the syringe device of Fig. 2C were used to obtain many individual aliquots from a certain level in a core. We take some of these aliquots and spike each with a single marker tablet. We anticipate that when an average marker tablet ($\bar{M} = 12,489$ spores) is combined with an average sediment aliquot ($\bar{V} = 0.2071 \text{ cm}^3$) the resultant ratio $M/V = 60,300 \text{ markers/cm}^3$. Now imagine that we obtain more

sediment samples by combining 4 aliquots chosen at random and add 4 marker tablets to each sample. Again we anticipate that when an average spike ($\bar{M} = 49,956$) is combined with an average 4-aliquot sample ($\bar{V} = 0.8284 \text{ cm}^3$) the resulting ratio M/V is also 60,300. But would the ratios M/V really be constant in these replicate samples? If not, how would they vary?

Computer algorithms are available to generate normally distributed random numbers with specified mean and standard deviation. An algorithm was used to generate 200 artificial sediment aliquots with mean 0.2071 cm^3 and standard deviation 0.0037 cm^3 and 200 simulated marker tablets with mean 12,489 grains and standard deviation 491. These were combined at random, and the marker grains/ cm^3 in the 200 samples are plotted in the upper part of Fig.3. A similar procedure was used to obtain M/V in the 200 simulated 4-aliquot samples shown at the bottom of the figure.

The analyst who assumes that M/V in his samples is a constant is simply wrong. I show the zero point for the graph scales of Fig.3 to stress how much variability there really is. We can reduce the scatter of M/V by using multiple-aliquot samples, but M/V is still variable. Although the pollen analyst is more interested in the samples' fossil pollen than in their markers, we will see that little can be known about the pollen without understanding the variable M/V . The concepts necessary for that understanding will be useful later when the concentration of the fossil grains is discussed.

We need to predict the scatter of M/V in replicate samples without going to the bother of simulated trials. As a geologist, I find it difficult to deal with abstract mathematics unless I can relate them to a model that is real or at least can be pictured in the mind.

Let us assume that the number of markers per tablet and the sediment volume per aliquot both follow normal distributions. How can we compare

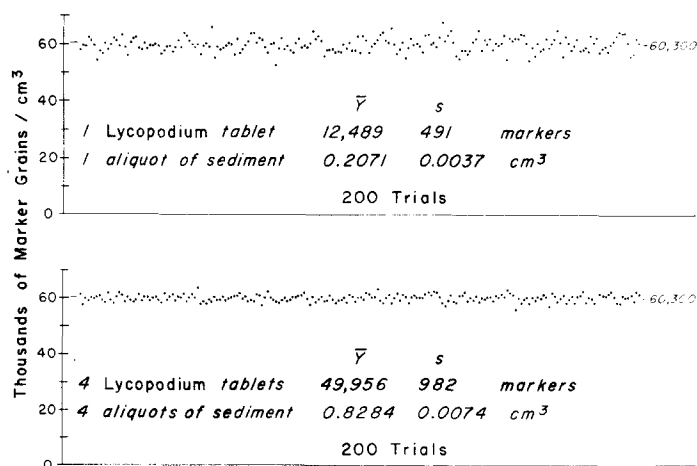


Fig.3. Analysts assume that the number of marker grains/ cm^3 in their samples is constant. Computer simulations show this assumption is false.

these two distributions to examine the possible values M/V can take if we select a marker tablet at random and add it to a random sediment aliquot? Because of the very different means and variances of the two distributions, it is convenient to standardize the two populations so that they can be plotted in units of their respective standard deviations. This can be done for any value of a variable by subtracting the population's mean and dividing the result by the population's standard deviation. The resultant curve of standard normal probability will have a mean of zero and a standard deviation of one. The area under the standard normal density function is unity. Fig. 4 shows the distributions of the variables M and V coded and plotted in standard form in units of standard deviation which I will call Z units.

I emphasize that this is done for our convenience. Any Z unit can be transformed back into the original units by multiplying it by the population's standard deviation and adding its mean. The diagonal lines across the figure show selected values of the quotient M/V in the original populations. For example, if a marker tablet happened to contain exactly 12,489 grains and it were placed in a sediment aliquot of exactly 0.2071 cm^3 , then the sample's M/V ratio would equal 60,300, and the point would plot at the center, or origin, of the diagram where the two population means intersect at $Z = 0$. The figure makes it clear that many other possible combinations of M and V also have the same value 60,300, and that all these combinations plot along a straight line that passes through the center of the diagram. Other values

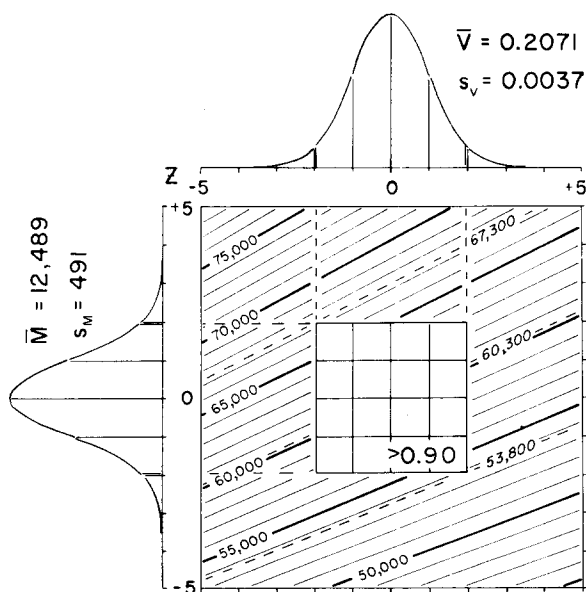


Fig. 4. The marker grains per tablet M and sediment volume per aliquot V are normally distributed variables. Their probability distributions have been coded with zero means and plotted in Z units of standard deviation. The diagonal lines connect points of constant M/V in the populations' original units.

greater than 60,300 will plot along straight lines in the upper part of the figure; those less will plot in the lower part of the figure. With all these possible combinations of M/V , what can the pollen analyst say about limiting values that will contain M/V at some stated probability?

In an early effort to assign limits (Maher, 1972, p.544), I assumed that if 0.95 of the values for M occurred within 1.96 standard deviation units of the population mean, and if 0.95 of the values for variable V occurred within 1.96 Z units of its mean, then there was a probability of $0.95 \times 0.95 = 0.90$ that both variables would simultaneously or jointly occur within their respective intervals. This range is shown by the central square box in Fig.4. The maximum limit of M/V that satisfies these criteria is at the upper left of the box where $M/V = 67,300$ markers/cm³. The minimum value of M/V at the lower right corner of the box is 53,800 markers/cm³. The interval from 53,800 to 67,300 also includes many other possible combinations of M/V that would result if one or both of the variables lay farther from their means than the stipulated 1.96 units. Therefore, the limits at the opposite corners of the box are labeled >0.90 limits. We may anticipate that they are much greater than 0.90.

A much better procedure to calculate the limits of M/V involves the notion of circular normal probability. Beyer (1968, pp.151–153) provides the normal joint probability density function:

$$f(x, y) = 1/(2\pi\sigma^2) \exp \{-(x^2 + y^2)/(2\sigma^2)\}$$

where x and y are two random variables, normally and independently distributed, each with mean zero and standard deviation σ . In the case of the standardized data in Fig.4, which are coded with zero means and unit standard deviations, the standardized bivariate normal density function $f(x, y) = 1/(2\pi) \exp \{-(x^2 + y^2)/2\}$. As the dimensions of the face of the graph (Fig.4) are in Z units, it is convenient to transform from rectangular (x, y) to polar (Z, θ) coordinates where Z is the distance from the origin. With the transformation, the joint probability density function $f(Z, \theta) = Z/(2\pi) \exp \{-Z^2/2\}$.

We may visualize this standard normal joint probability density function as defining a bell-shaped surface lying over the (x, y) plane (Fig.5). The height of the surface above any point on the (x, y) plane at distance Z from the origin is $1/(2\pi) \exp \{-Z^2/2\}$. The volume under the surface is proportional to the joint probability density (JPD) between the two variables, and as Z increases, the volume and JPD approach 1. The sum of the JPD which lies within a distance of Z from the origin $= 1 - \exp \{-Z^2/2\}$ (A. Gordon, pers. comm., 1978).

As is shown in Fig.6, slightly more than half of the total JPD occurs within 1.2 Z units of the origin. About 0.99 of the JPD lies over a circle with a radius of 3 Z units. Essentially all ($\Sigma p = 0.9998$) of the JPD will lie within 4.1 Z units of the origin. That is, when one marker tablet is added to a single aliquot of sediment, the marker concentration will lie somewhere between 50,000 and 71,000 grains/cm³. If we compare Fig.6 with the 200 random samples of M/V in the upper part of Fig.3, all the points fall within these limits.

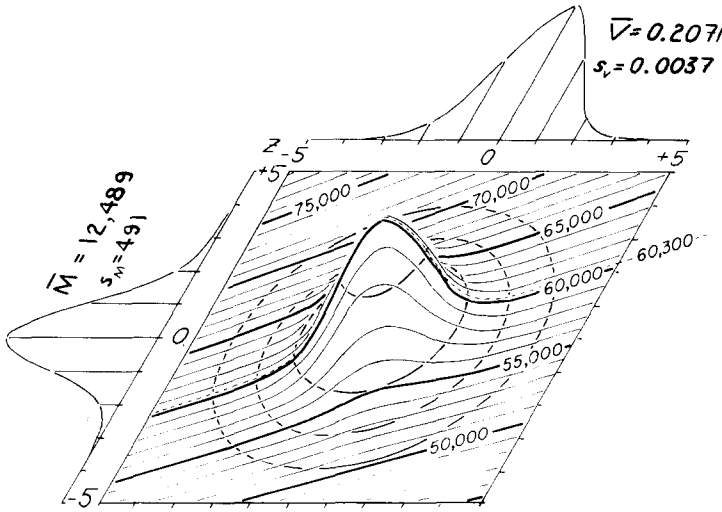


Fig.5. The standard normal joint probability density function defines a bell-shaped surface lying over the (x, y) plane. The volume under the surface is proportional to the joint probability density (JPD) between the variables. As Z increases, the JPD approaches 1.

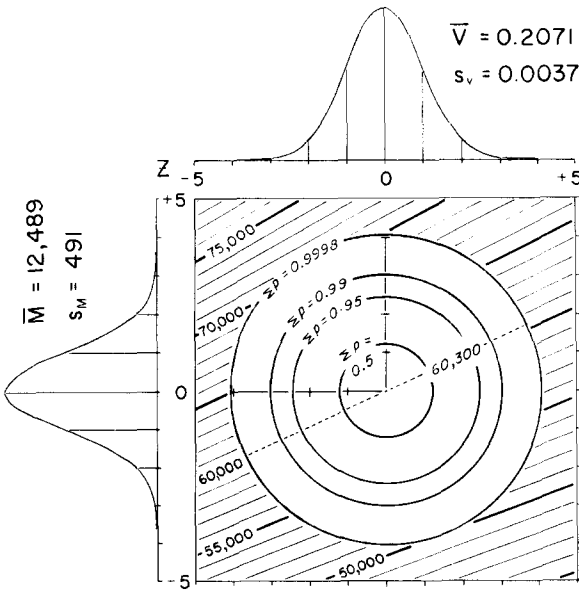


Fig.6. Essentially all ($\Sigma p = 0.9998$) of the variables' joint probability density (JPD) lies within 4.1 Z units of the origin.

But complete certainty in statistics or in life is not very interesting. After all, we could save time by concluding with absolute certainty that the number of markers/volume would lie somewhere between 0 and ∞ ; then completely accurate (and meaningless) pollen concentration diagrams could be drafted without ever looking at a pollen grain!

Fig.6 can be regarded as a plan view of the bell-shaped mound of JPD of the variables M and V ; the straight lines connecting the points where M/V is constant can be thought of as the position of planes which are at right angles to the plan view and which intersect and slice through the mound of JPD. We must now select the position of two such limiting planes so that the JPD contained between the slices is some convenient portion — say 0.95 or 0.99 — of the total.

The problem is illustrated on the generalized inset diagram in Fig.7. Assume that two vertical planes, representing an upper and lower limit of M/V , intersect the mound of JPD such that they are tangent to the basal circle C of radius Z . The JPD lying above the circle $C = 1 - \exp \{-Z^2/2\}$. But additional JPD also occurs over that part of the outer annulus which lies outside circle C , but inside the limiting planes, or over the area labeled A . Again, because essentially all ($\Sigma p = 0.9998$) of the JPD occurs within 4.1 Z units of the origin, for the following discussion we can ignore the trivial volume lying at a greater distance from the origin.

Because of the symmetry of the bell-shaped mound, it can be seen that for any value of Z we may choose, the two portions of JPD lying over that part of the outer annulus outside the limiting planes are always equal. That is to say, we may expect to calculate the total JPD overlying circle C and that part (A) of the outer annulus between the limiting planes by specifying only the radius of the circle C to which the planes are tangent; we need not know the

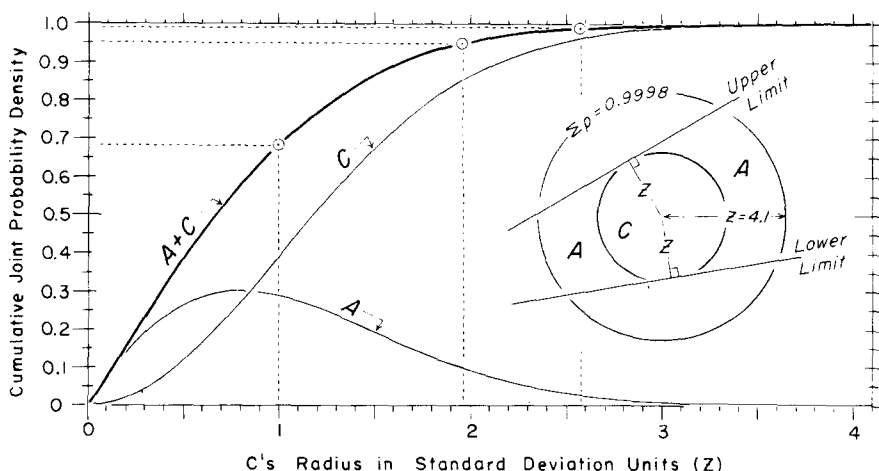


Fig.7. The volume of the joint probability density (JPD) between the M/V limits has the same numerical value as the area under the standard normal density function within $\pm Z$ units of the mean.

precise angle between the limiting planes so long as they do not intersect within $4.1 Z$ units of the origin. Fig. 7 illustrates how the JPD over areas C and A changes as the radius of circle C increases from 0 to $4.1 Z$ units. When the radius of C is small, very little JPD lies over it, but the volume over A increases rapidly. As C's radius gets larger, more volume lies over it, and the JPD over A decreases. We are, of course, interested in how the sum of the JPD over both C and A changes as Z increases, and this is shown by the heavy curve labeled A + C. Notice that when the limiting planes are tangent to a circle C of radius $Z = 1$, about 0.68 of the JPD lies between the limits. When $Z = 1.96$, 0.95 of the density lies between the limits; and when $Z = 2.576$, 0.99 of the JPD is contained between the limits. In fact, the volume between the limiting planes has the same numerical value as the area under the standard normal density function within $\pm Z$ units of the mean, for which tables are readily available (e.g. Rohlf and Sokal, 1969, pp.157–158).

If we wish to predict limits that would be expected to contain the actual marker grains/cm³ in about 95 of 100 sample trials — that is, if we wish to set 0.95 confidence limits for the true number of markers in a cm³ of sediment resulting from adding one marker tablet to a single aliquot of sediment — we would need to set limiting planes tangent to a basal circle with a radius of $Z = 1.96$. Such a circle would just fit inscribed into the central square of Fig. 4 (see also Figs. 5 and 6). By interpolating where lines of constant M/V graze the circle, we may estimate that the lower and upper 0.95 confidence limits would be approximately 55,000 and 65,000 markers/cm³.

I have used these figures to illustrate the problem and to suggest how limits may be obtained. Obviously we do not want to draw such figures if the answers can be calculated directly. It may already have become apparent that the labels of the slanting lines which connect points of constant M/V are really slopes of lines relating marker grains to sediment volume. A slope measured in tens of thousands implies a line whose rise per run is nearly vertical. That the slopes of the lines of constant M/V in the figures (e.g. Fig. 4) are not vertical results from the fact that these data have been scaled in units of standard deviation which also scale the slopes of the lines. Close examination will show that the lines are not parallel but appear to radiate from some point of origin far to the lower left of the diagram. The situation is shown in Fig. 8. The location of this zero point from which the lines radiate can be found easily if we recall that the figure is scaled in units of standard deviation or Z . The center or origin of the graphs we have been discussing (e.g. Fig. 6, see also inset at left of Fig. 8) is $\bar{M}/s_m = 12,489/491 = 25.4 Z$ units above and $\bar{V}/s_v = 0.2071/0.0037 = 56.0 Z$ units to the right of this zero point. These two sides form a right triangle with an hypotenuse of $61.5 Z$ units. The slope of the hypotenuse or angle $\alpha = \arctan \{(\bar{M}/s_m)/(\bar{V}/s_v)\} = 24.44^\circ$. This scaled slope can be decoded back to the original units of $\bar{M}/\bar{V}$ from the following: $\bar{M}/\bar{V} = (s_m \tan \alpha)/s_v = 60,300 \text{ grains/cm}^3$.

The two limiting lines that are tangent to the circle of radius Z radiate from the zero point on either side of the hypotenuse, and they differ from α by the angle β . It is apparent that $\sin \beta = Z/\text{hypotenuse}$ so that $\beta = \arcsin$

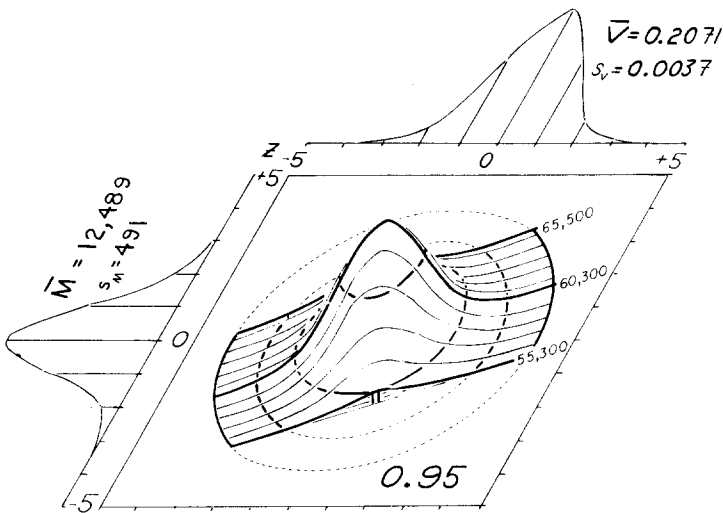


Fig.9. Limiting values of M/V which contain 0.95 of the joint probability density (JPD) for the variables M and V .

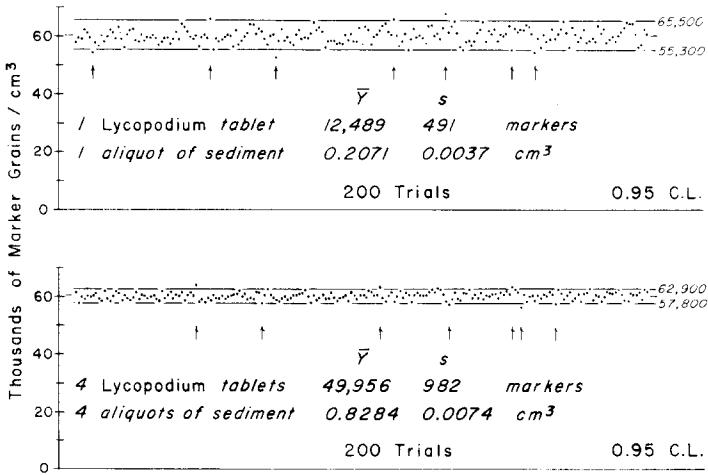


Fig.10. 0.95 confidence limits for the true value of M/V in samples composed of single marker tablets and volume aliquots are shown at the top. The limits at the bottom are for samples containing four marker tablets and four volume aliquots. The 400 synthetic trials are from Fig.3. Arrows point to samples which lie outside the limits.

INDIGENOUS GRAINS/MARKERS: THE VARIABLE R

We have examined the probability distributions of the variables M , V , and M/V . The only other variable in the calculation of pollen concentration by the spike technique is R , which is a function of the true ratio u of indigenous grains to marker grains in the sample. This is the only property of the

individual sample that is subject to direct measurement. If all indigenous and marker grains were counted, we would know the true value of u . However, the pollen analyst does not want to count every grain in the preparation. Enough grains and markers are identified to obtain an estimate of u based on the average ratio $\bar{R}$ in the count; $\bar{R} = x/n$ which is $\hat{u}$, the point estimate of the sample's true u .

The time at the microscope is spent to get an estimate of the possible values of the ratio of indigenous pollen to marker grains in the sample in order to make a reasonable judgment of the true ratio. For example, if one counts 50 indigenous grains while tallying 300 marker grains, one has accumulated 300 samples of the number of indigenous grains per single marker grain, 150 samples of indigenous grains per 2 markers, 100 samples of indigenous grains per 3 markers . . . down to one sample of 50 grains per 300 markers. The mean of each sample group will be 0.1667, of course, but the deviation of the samples about the mean decreases as the pollen grains are averaged over more and more markers. Mosimann's equation for calculating confidence limits of u requires only that one tabulates the indigenous grains x while counting up to a fairly large number of markers n . The equation (Mosimann, 1965, p.659) may be written as follows:

$$u_L = \frac{\hat{u} + [Z^2/(2n)] \pm Z \{[(\hat{u}(1+\hat{u})/n) + [Z^2/(4n^2)]]^{0.5}}{1 - (Z^2/n)}$$

where $\hat{u} = x/n$ and Z is the standard deviation unit to include a specified area under the standard normal curve. For 0.95 confidence limits of u , $Z = 1.960$; for 0.99 limits, $Z = 2.576$, etc.

The pollen count contains the information needed to reconstruct the frequency distribution of the variable R , the indigenous/marker ratio in the single sample. This distribution can be reconstructed by solving Mosimann's equation substituting Z 's 1, 2, 3, etc. The distribution is positively skewed with the upper limits farther from $\hat{u}$ than are the lower limits. This was also noted in the distribution of the variable M/V . If the variables M/V and R are transformed to $\log(M/V)$ and $\log R$ their distributions will assume a log-normal character. Mosimann and Greenstreet (1971) used a logarithmic transformation of pollen ratios in their search for representation-insensitive methods to interpret pollen sequences (see also Maher, 1972, fig.6, p.549).

CONFIDENCE LIMITS FOR POLLEN CONCENTRATION

We have been considering distributions of variables and the probability statements that can be made about them. It is now time to define some parameters that will be useful in assigning confidence limits to pollen concentration. Let γ represent the true number of indigenous pollen grains of a taxon X per cm^3 of sediment whether in a core or in a sample extracted from one level in the core.

For an individual sample from one level in a core (γ in sample is an unknown constant): v = true volume of the individual sediment sample in

cm^3 ; μ = true number of marker grains in the individual sample; and ρ = true ratio of the number of indigenous pollen grains to marker grains in the individual sample (Note: ρ is the same as u). These parameters differ in value for each sample taken from a single level, but they are true parameters for the individual sample. Thus, $\gamma \equiv \rho\mu/\nu$.

For a large number of replicate samples from one level in a core (γ assumed constant): λ = true mean of sample volumes (cm^3) in a large number N of samples ($\lambda = \Sigma\nu/N$); η = true mean of marker grains per sample in a large number N of samples where the same number of tablets were added to each sample ($\eta = \Sigma\mu/N$); η/λ = true mean ratio of marker grains/ cm^3 in a large number N of samples:

$$\eta/\lambda = \left[\prod_{i=1}^N (\mu/\nu)_i \right]^{1/N}$$

(that is, η/λ is the geometric mean of the (μ/ν) 's); ψ = true mean ratio of number of indigenous pollen grains to marker grains in a large number N of samples:

$$\psi = \left[\prod_{i=1}^N \rho_i \right]^{1/N}$$

$\gamma \equiv \psi\eta/\lambda$, or in terms of the individual samples' parameters:

$$\gamma = \left[\prod_{i=1}^N \rho_i \right]^{1/N} \times \left[\prod_{i=1}^N (\mu/\nu)_i \right]^{1/N}.$$

I find the following mental exercise helpful. Assume that the pollen concentration in sediment from a certain level in a core is γ , an unknown constant. Drop a single marker tablet into a single volume aliquot of the sediment and mix. At that instant three additional unknown parameters are fixed. The sample has (1) a discrete number of markers μ , which represents the value that the variable M happened to take in that particular tablet, and (2) a discrete sediment volume ν , which represents the value that the variable V happened to take in that volume aliquot. Hence, the sample has a fixed value of μ/ν , which represents the value that the variable M/V happened to take in the sample. But because the sediment's pollen concentration γ is a constant, and $\gamma \equiv \rho(\mu/\nu)$, the inverse relationship (correlation coefficient = -1) between μ/ν and ρ means that (3) the act of adding the marker tablet to the sediment fixes that sample's parameter ρ , its true ratio of indigenous pollen to marker grains. That is, we have fixed μ and ν (hence, μ/ν) and ρ .

If we imagine all possible combinations of (μ/ν) 's in N such samples, they will group around η/λ forming a distribution of parameters that is skewed to the right. This grouping can be thought of as a probability distribution of the variable M/V or of all possible (μ/ν) 's that compose it. The important point is that for each sample μ/ν there is also a discrete sample ρ . All possible ρ 's will

form a grouping that is — because of the inverse relationship between ρ and (μ/ν) — an exact mirror image of the distribution of the (μ/ν) 's. That is why (η/λ) must be the geometric mean of the (μ/ν) 's and ψ must be the geometric mean of the ρ 's so that the product of η/λ and ψ will equal γ .

I have said that a logarithmic transformation of the variable M/V restores the distribution to one that is normal. For the moment I will assume this to be true; later I will discuss the range of conditions under which the assumption is valid. I will also assume that the distributions from which the marker tablets and sediment volume aliquots are drawn are normal and that their parametric means and variances are known from previous experiments. The log transformation of the variable M/V , then, results in a normal distribution with mean $\log(\eta/\lambda)$ and a variance we will call δ^2 ; that is, $\log(M/V) \sim N[\log(\eta/\lambda), \delta^2]$. This spaces the $\log(\mu/\nu)$'s normally about $\log(\eta/\lambda)$ and simultaneously spaces the “phantom distribution” of $\log \rho$'s normally about $\log \psi$. The variance δ^2 of the distribution of the random variable $\log(M/V)$ is also the variance of the distribution of the $\log \rho$'s about $\log \psi$. As the ρ 's are the true indigenous/marker ratios in the many potential individual samples, we can regard the $\log \rho$'s as a distribution of the variable $\log R_{\text{true}}$; it follows that $\log R_{\text{true}} \sim N[\log \psi, \delta^2]$.

As a pollen analyst I would like to get an estimate of pollen concentration γ by studying only one sediment sample. When I pick up the vial of processed pollen made from this sample, I know it has an unknown parameter $\log \rho$ which is the logarithm of the true ratio of indigenous pollen to marker grains in the vial. Further, given sediment of concentration $\log \gamma$, I know the sample's $\log \rho$ is completely dependent on the sample's parameter $\log(\mu/\nu)$ because $\log \rho \equiv \log \gamma - \log(\mu/\nu)$. I can learn this sample's $\log \rho$ if I count all indigenous grains and all marker grains in the vial and take the logarithm of their ratio. As that is impractical, I decide to count only a representative portion of the population on a microscope slide. I may infer from Mosimann's equation that if I counted many subsamples in this slide (or slides), each comprising a set number of marker grains, these indigenous/marker ratios (R_{count}) would have a lognormal distribution where $\log R_{\text{count}} \sim N[\log \rho, \omega^2]$. The variables $\log R_{\text{true}}$ and $\log R_{\text{count}}$ belong to distributions with different means and variances; the variables are normally and independently distributed. The count variance ω^2 must be added to the variance δ^2 which was conditional on the variability of the marker tablets and volume aliquots. Therefore the unconditional distribution of $\log R \sim N[\log \psi, \delta^2 + \omega^2]$.

Now it is the pollen analyst who decides the value that is to be taken by the marker grain number n ; whether it is 1 or 10,000, it is the analyst who stops counting and fixes n . And it is here where the Work Ethic comes in: the more counting work is done, the smaller ω^2 becomes. The variance δ^2 is built into the problem from the time the marker tablet(s) is added to the sediment. The variance ω^2 depends entirely on the numerical values of x and n in the final pollen count. We can make ω^2 approach zero with huge expenditures of effort; we cannot reduce δ^2 once the sediment is spiked.

The unconditional distribution of $\log R$ can be given meaning in terms of

indigenous grains by supplying information about the possible distribution of marker grains/cm³ in the sample: $\log (M/V) \sim N[\log (\eta/\lambda), \delta^2]$. This can be done by adding $\log (\eta/\lambda)$ to the parametric mean $\log \psi$ of the unconditional distribution of $\log R$. It follows from:

$$\log R \sim N[\log \psi, \delta^2 + \omega^2]$$

that:

$$\log R + \log (M/V) \sim N[\log \psi + \log (\eta/\lambda), \delta^2 + \omega^2]$$

which by definition equals:

$$\log R + \log (M/V) \sim N[\log \gamma, \delta^2 + \omega^2]$$

We can regard $\log R + \log (M/V)$ to be a random variable which is distributed normally about mean $\log \gamma$ with population variance $\delta^2 + \omega^2$. Part of the variance, δ^2 , arises from the variability of the marker grains and sediment aliquots; the rest, ω^2 , is related to the numerical values of x and n in the pollen count.

Assume that we added one marker tablet to an aliquot of sediment, processed the sample and counted 300 marker grains while identifying 50 indigenous grains. Let us try to assign 0.95 confidence limits to the true pollen concentration in the sediment. We will need to derive estimates of the parameters $\log \gamma$, δ^2 , and ω^2 . The only reasonable estimator for γ is $\bar{R}\bar{M}/\bar{V} = (50/300)12,489/0.2071 = 10,050$ grains/cm³ which the pollen analyst normally uses. This estimate would only be exactly right in the rare instance where the sample's μ/ν really equaled η/λ so that the sample's $\rho = \psi$; further, the mean indigenous/marker ratio x/n in the count would also have to be exactly ρ . I am implying that the estimate of γ is not very good. The point of this exercise is to find how poor it really is.

Fig.11 may help to visualize the problem. The curve shown along the y-axis represents the variable $\log (M/V)$. $\log (M/V) \sim N[\log (\eta/\lambda), \delta^2]$. The variance δ^2 of this distribution is controlled by the variability of the marker and sediment aliquots. The distribution serves as a proxy for $\log R_{\text{true}} \sim N[\log \psi, \delta^2]$ in a large number of samples, and it furnishes part of the total variance of the ratio x/n in the single sample's count. Our estimator of $\log (\eta/\lambda)$ is $\log (\bar{M}/\bar{V})$. An estimate of δ^2 can be obtained by using the procedure described in Fig.8 to set probability limits on the variable M/V . We can compute the limits that contain 0.6826 of the JPD for M/V by setting Z equal to 1; this will affect angle β . The upper limit $(M/V)_{\text{max}} = 62,928$ markers/cm³, and the lower limit $(M/V)_{\text{min}} = 57,719$ markers/cm³. $\log (M/V)$ is essentially normally distributed, and the log limits will be situated at one standard deviation from the log mean. If the distribution is exactly lognormal, half the difference between the log limits will equal $s_{\log (M/V)}$, an estimator for δ . That is, $\overline{\log (M/V)} = \log (\bar{M}/\bar{V}) = 4.78035$, and $s_{\log (M/V)} = (\log 62,928 - \log 57,719)/2 = 0.01876$. The distribution of $\log (M/V)$ in Fig.11 has been standardized or coded with zero mean and unit variance.

The remaining variance ω^2 that affects the ratio x/n in the single sample

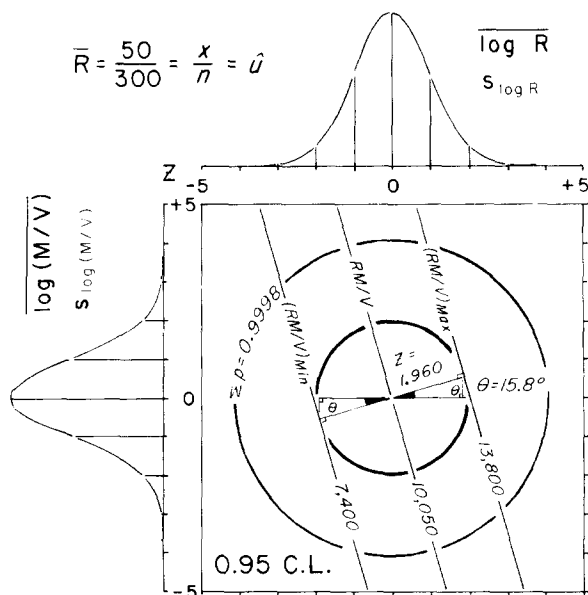


Fig.11. 0.95 confidence limits for the true concentration of a taxon X in a hypothetical sample. See text for discussion.

results from the size of x and n in the pollen count. $\log R_{\text{count}} \sim N[\log \rho, \omega^2]$. If we solve Mosimann's equation with $x = 50$, $n = 300$, and $Z = 1$, we obtain ratio limits ($R_{\max} = 0.19450$ and $R_{\min} = 0.14330$) whose logarithms have a normal spacing about $\log(x/n)$; we will call this $\log \bar{R}$, and use it as an estimator for $\log \rho$. Half the difference of the log limits is $s_{\log R}$, an estimator for ω . In Fig.11, $\log \bar{R} = \log(50/300) = -0.77815$, and $s_{\log R} = (\log 0.1945 - \log 0.1433)/2 = 0.06634$. Again, this distribution is standardized in Z units.

All points of equal $\log R + \log(M/V)$ will plot as lines across the figure, and all such lines are parallel. We can predict that the scaled gradient from lower to higher values of $\log R + \log(M/V)$ will differ from a horizontal line passing through the origin by θ . ($\theta = \arctan [s_{\log(M/V)}/s_{\log R}] = 15.8^\circ$.) The lines tangent to a basal circle of 1.96 Z units on this diagram contain between them 0.95 of the JPD of the variable $\log R + \log(M/V)$. We need to know the lines' values. It can be seen that these points of tangency contain components from both distributions. For distribution $\log(M/V)$ these values are situated at $Z \sin \theta = 0.53$ standard deviation units on either side of the mean; for distribution $\log R$, the corresponding values are $\pm Z \cos \theta = \pm 1.89$ Z units with respect to the mean. These Z units can be decoded to $\log(M/V)$ and $\log R$ by multiplying them by their respective distributions' standard deviations and adding their means.

By factoring and regrouping:

$$\log \text{ limits} = \overline{\log(M/V)} + \overline{\log R} \pm Z(s_{\log(M/V)} \sin \theta + s_{\log R} \cos \theta)$$

Because the term:

$$(s_{\log(M/V)} \sin \theta + s_{\log R} \cos \theta) = [(s_{\log(M/V)})^2 + (s_{\log R})^2]^{0.5}$$

the 0.95 confidence limits for the true concentration of indigenous pollen can be expressed as:

$$\log \text{ limits} = \overline{\log(M/V)} + \overline{\log R} \pm 1.96[(s_{\log(M/V)})^2 + (s_{\log R})^2]^{0.5}$$

In Fig. 11, $\log L_1 = 3.87$ and $\log L_2 = 4.14$. L_1 and L_2 , respectively, are approximately 7400 and 13,800 indigenous pollen grains/cm³.

We can take this relationship out of the logarithmic domain by defining:

$$F = \text{antilog}\{1.96[(s_{\log(M/V)})^2 + (s_{\log R})^2]^{0.5}\}$$

L_1 is our estimate of pollen concentration, $\overline{RM}/\overline{V}$, divided by the term F , and L_2 is $\overline{RM}/\overline{V}$ multiplied by F . The probability that $(\overline{RM}/\overline{V})/F \leq \gamma \leq (\overline{RM}/\overline{V}) \times F = 0.95$. We can be reasonably certain (0.95 certain) that the interval between L_1 and L_2 includes γ , the true pollen concentration in the sediment. This is not to say that there is a probability of 0.95 that γ is really in this particular interval, for it is either in or out. Rather we should interpret this probability statement to mean that on the average we expect 95 of 100 such intervals similarly obtained from single sediment samples to include the true number of indigenous pollen grains/cm³.

A TEST OF THE METHOD

We could test this procedure for calculating confidence intervals by taking many samples from homogeneous sediment and calculating the $\overline{RM}/\overline{V}$'s. If 0.95 confidence limits were calculated for each sample, we could see whether 95% of them would cover γ . A much easier approach would be to simulate the problem with a computer.

Let us assume a hypothetical sediment that contains $\gamma = 120,600$ indigenous pollen grains/cm³. Assume also that we will take one marker tablet drawn at random from a distribution with mean 12,489 and standard deviation 491; we will add it to a single aliquot of sediment drawn with a device that takes sediment samples with mean 0.2071 cm³ and standard deviation 0.0037. An algorithm to produce random numbers that form a normal population with a specified mean and standard deviation generates the synthetic marker tablets and sediment aliquots which are then combined.

One such sample will illustrate the process. The computer assigned the random marker variable M the value 11,955 which represents μ , the true number of marker grains in the individual sample. The sediment volume variable V was assigned the value 0.2063 cm³ which is ν , the true volume of sediment in the sample. The random variable M/V in this sample thus takes the parametric value $\mu/\nu = 57,950$. From the definition of pollen concentration, $\gamma \equiv \rho(\mu/\nu)$, we have $120,600 = \rho(57,950)$, so ρ in this particular sample is 2.0811. In this case ρ is known because we have knowledge about the sample not usually available. An analyst handed this sample could establish its ρ with certainty only by counting its combined total of 36,835 pollen and

marker grains! (Notice that even with this Herculean effort, his concentration estimator $\overline{RM}/\overline{V}$ would yield $2.0811(12,489/0.2071) = 125,500$ grains/cm³. This estimate is 4% greater than γ .)

The computer next needs to simulate a pollen count that an analyst might make on this sample. We know that the possible indigenous/marker ratios in the counts will be distributed about ρ and that $\log R_{\text{count}} \sim N[\log \rho, \omega^2]$. The variance ω^2 is a function of the size of the counts. I specify that the simulated analyst will count x pollen grains until $n = 300$ marker grains have been tabulated. Because this sample's $\rho = 2.0811 = x/300$, $x = 624$ (rounded to the nearest whole grain), and $\log \rho = \log(624/300) = 0.31806$. Mosimann's equation for 0.6826 confidence limits ($Z = 1$) was solved for $x = 624$ and $n = 300$. The limits were converted to logarithms, and half the difference of the log limits provides $s_{\log R} = 0.03054$ which is an estimator for ω . The computer then generates a random value from a normal population with mean 0.31806 and standard deviation 0.03054. It happened to be 0.29646, the antilog of which is the ratio 1.979. The computer multiplies this $\hat{u}$ by $n = 300$ to obtain $x = 594$ (rounded to the nearest whole grain). This synthetic pollen count of 594 pollen grains and 300 marker grains is a realistic estimate of an actual count we might have measured on this synthetic sample. The computer then uses this pollen count to calculate 0.95 confidence limits for γ . In this case $L_1 = 101,500$, and $L_2 = 140,500$; the interval easily includes the sediment's defined pollen concentration of 120,600 grains/cm³.

The sample we have been discussing is shown as the first sample in the upper part of Fig.12. A similar procedure was used to produce the other 199 sample trials to its right. The 200 intervals in the lower part of Fig.12 are synthetic trials involving samples combining four aliquots of sediment with four marker tablets and extending the n count to 1200. These latter confidence intervals are a good deal smaller, as might be expected. The success rate in the upper series of 200 trials was 0.96, and that in the lower series was 0.925. A later computer simulation of each of the two sampling strategies was carried out through 5000 trials. Both 0.95 limits ($Z = 1.96$) and 0.99 limits ($Z = 2.576$) were calculated. The success rates in these trials, rounded to two decimal places, were 0.95 and 0.99.

The method of assigning confidence limits to pollen concentration based on a single sample seems to be functioning as expected. I was interested to see how the 0.95 limits compare to the >0.86 limits I have previously used (Fig.13). The comparison shows that the >0.86 limits were really ≥ 0.86 limits.

RESTRICTIONS ON THE TECHNIQUE

So far we have assumed that the variables $\log(M/V)$ and $\log R$ are distributed normally. It is now necessary to consider the range of conditions under which these assumptions are reasonable. I have used $\log(\overline{M}/\overline{V})$ as an estimator for $\log(\eta/\lambda)$. My $s_{\log(M/V)}$, an estimator for δ , is determined by first calculating at $Z = 1$ the limits which contain 0.6826 of the JPD for the variable M/V .

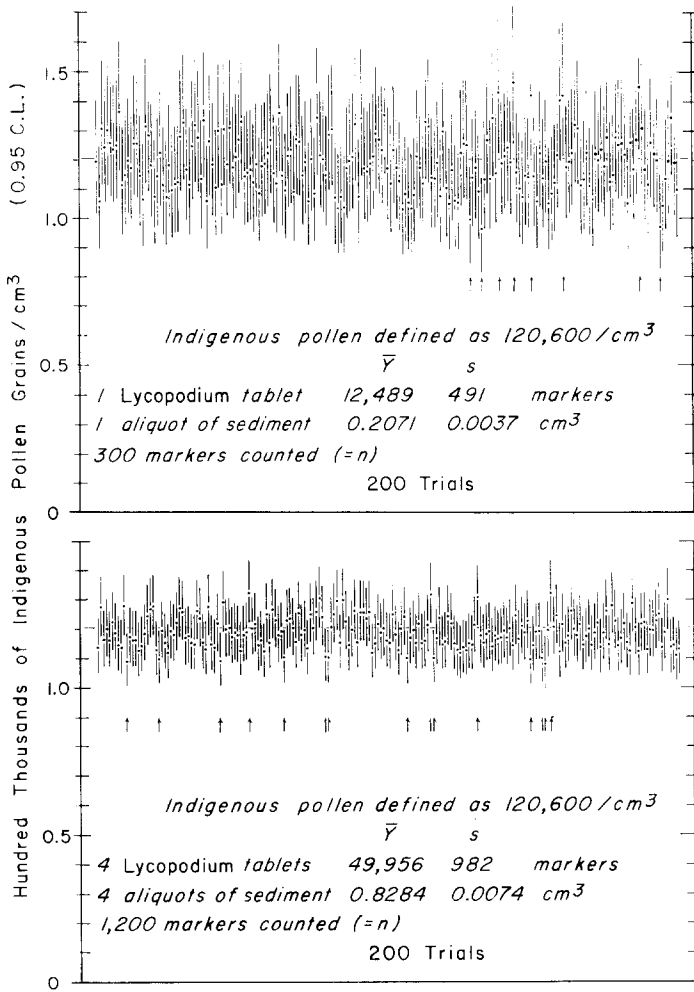


Fig.12. 0.95 confidence intervals for the true concentration of indigenous pollen/cm³ for synthetic samples and counts that were generated with a computer. Arrows point to intervals which do not contain the sediment's defined pollen concentration.

These limits are converted to logarithms, and half their difference is $s_{\log(M/V)}$. If the distribution is exactly normal, $\log(M/V)$ will be equidistant between the log limits. These comments also apply to $\log \bar{R} = \log(x/n)$, an estimator of $\log \rho$, and $s_{\log R}$, an estimator of ω . My $s_{\log R}$ is determined by halving the total interval between the logarithms of the limits determined by solving Mosimann's equation with $Z = 1$. Again, if the distribution is exactly normal, $\log \bar{R}$ will be equidistant between the log limits.

Now consider a case where the variables' distributions were not normal. When we make the simplifying assumption that they were normal and use $\log(M/V) \pm Z(s_{\log(M/V)})$ and $\log \bar{R} \pm Z(s_{\log R})$ to make probability statements about the variables $\log(M/V)$ and $\log R$, we must realize that the disagreement

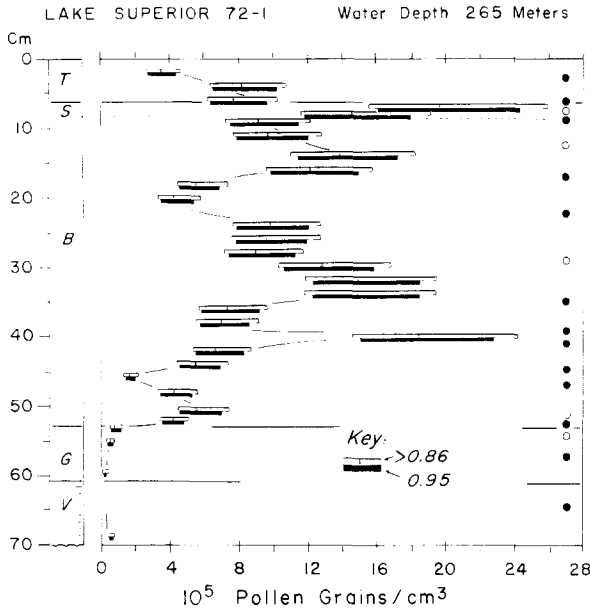


Fig.13. An abbreviated concentration diagram from Lake Superior (Maher, 1977, fig.4, p.32) comparing the original >0.86 confidence intervals with the 0.95 intervals of this paper. The sediment from the base upward includes: varved clay, gray clay, brown silty clay, material since settlement, and tailings. The round symbols at the right margin indicate depth levels where the null hypothesis that the pollen concentration in adjacent samples is the same is rejected at the $p = 0.05$ level of confidence; the solid dots indicate rejection also at the $p = 0.01$ level. The basis for the null test is discussed later in the text.

between the predicted limits and the actual limits will increase at greater Z distances. The question thus resolves itself to asking how much difference — at some arbitrarily large Z distance from the distribution's mean — one is willing to accept between the exact limit and that limit predicted by assuming the log variables have a normal distribution.

I now make the relatively stringent requirement that before I accept the hypothesis that the log variables are normally distributed, neither of the anti-logs of the upper or lower limits of the solutions obtained by assuming the normal distribution of $\log(M/V)$ and $\log R$ will — at $+4$ or -4 Z units from the mean — differ by more than 10% from the values calculated by an exact solution. For the exact solution of the bounding limits of the interval which contains 0.99994 of the JPD of M/V , I redefine angle β making $Z = 4$ rather than 1; that is:

$$\beta = \arcsin \{4/[(\bar{M}/s_m)^2 + (\bar{V}/s_v)^2]^{0.5}\}$$

In a similar manner, I obtain the “exact” bounding limits of the interval which contains 0.99994 of the probability of R by solving the Mosimann equation with $Z = 4$.

The variable $\log(M/V)$ meets this test of having an acceptably normal

distribution so long as the coefficients of variation ($100s/\bar{Y}$) of the parent distributions of M and V are both less than 9.1%. Put another way, neither $\bar{M}/s_m$ nor $\bar{V}/s_v$ should be less than 11. The restriction should rarely pose a problem for a careful worker. In order that the distribution of R be such that $\log R$ is acceptably normal in the sense mentioned above, the number of marker grains counted should be at least 100. The only other restriction is that for any value of n , there is a minimum value of x which should always be equalled or exceeded if acceptable normality of $\log R$ is to be maintained. When $n = 100$, confidence limits for concentration should be restricted to taxa with 12 or more grains; when $n = 200$, x should be ≥ 9 , and when $n \geq 400$, x should be ≥ 8 . It would seem prudent that these restrictions be observed when — as will be discussed later — two or more samples are combined or when null tests are made.

Mosimann's equation for assigning confidence limits to u relies on the fact that the negative binomial distribution approximates a lognormal distribution when n is fairly large. When dealing with relatively low values of x and n , it would be better to test the results predicted from the assumed lognormal distribution directly against those obtained from the negative binomial distribution itself. Mainland et al. (1956) have published tables of binomial confidence limits for percentages (or proportions based on counts made inside the sum) which are reproduced in most texts (e.g. Rohlf and Sokal, 1969, pp.208–214). When $\hat{p} = 0.04$, and the sum of pollen and markers is 100, the 0.95 limits of p are 0.0110 and 0.0993 (0.99 limits: 0.0068 and 0.1208). Now a proportion p based on an inside count can always be converted to a ratio u for an outside count by the equation: $u = p/(1 - p)$. Therefore, $\hat{p}$ and the binomial limits of p can be easily converted to $\hat{u}$ and the negative binomial limits of u . Here $\hat{u} = 4/96 = 0.04167$. The 0.95 limits of u are 0.0111 and 0.1102 (0.99 limits: 0.0068 and 0.1374).

If we visualize many samples in which x pollen grains were counted while tallying 96 marker grains, we would expect the x 's in 95% of the counts would range between 2 and 10, and that in 99% of the counts the x 's would range between 1 and 13. (Multiply the limits by 96 to solve for x , and recall that both x and n must be integers.) The prediction based on the lognormal assumption is that 95% of the x 's would fall between 2 and 10, and 99% would range between 2 and 14. The agreement is really quite good. I suggest that, as a rule of thumb, useful confidence limits can be assigned when $n \geq 100$ and $x \geq 4$. If "0.95" limits are assigned to taxa with three or fewer grains, be warned that they will have less than a 0.95 probability of including the taxon's γ .

AUTOMATING COMPUTATIONS

The steps used to compute confidence limits for the true concentration of a taxon are summarized in Table I. It is instructive and rather entertaining to solve a few confidence limit problems with pencil and paper. Then when the analyst has had all the fun he can stand, it is time to automate the solutions for reasons both of speed and accuracy.

TABLE I

Confidence limits for microfossil concentration measurements employing samples spiked with marker grains

- Given:
1. N_1 marker-grain aliquots (tablets) with known mean $\bar{Y}_1$ and standard deviation s_1 .
 2. N_2 sediment volume aliquots with known mean $\bar{Y}_2$ and standard deviation s_2 .
 3. Ratio (R) of number of grains of taxon X (x) to number of marker grains (n) counted in the sample. $\bar{R} = \hat{u} = x/n$.

$$\begin{aligned}\bar{M} &= N_1 \bar{Y}_1 &&= \text{mean number of marker grains added to sample} \\ s_m &= \sqrt{N_1} \cdot s_1 &&= \text{standard deviation of marker grains added to sample} \\ \bar{V} &= N_2 \bar{Y}_2 &&= \text{mean sediment volume in sample} \\ s_v &= \sqrt{N_2} \cdot s_2 &&= \text{standard deviation of sediment volume in sample}\end{aligned}$$

$$\alpha = \arctan \{(\bar{M}/s_m)/(\bar{V}/s_v)\} \quad (\text{Note: } \bar{M}/s_m \text{ and } \bar{V}/s_v \text{ should both be } \geq 11)$$

$$\beta = \arcsin \{1/[(\bar{M}/s_m)^2 + (\bar{V}/s_v)^2]^{0.5}\}$$

$$(M/V)_{\max} = [s_m \tan(\alpha + \beta)]/s_v$$

$$(M/V)_{\min} = [s_m \tan(\alpha - \beta)]/s_v$$

$$\bar{R} = \hat{u} = x/n \quad (\text{Note: } x \text{ should be } \geq 4 \text{ and } n \text{ should be } \geq 100)$$

$$R_{\max} = \frac{\hat{u} + [1/(2n)] + \{[\hat{u}(1 + \hat{u})/n] + [1/(4n^2)]\}^{0.5}}{1 - (1/n)}$$

$$R_{\min} = \frac{\hat{u} + [1/(2n)] - \{[\hat{u}(1 + \hat{u})/n] + [1/(4n^2)]\}^{0.5}}{1 - (1/n)}$$

$$\log \bar{R} = \log (x/n)$$

$$s_{\log R} = (\log R_{\max} - \log R_{\min})/2$$

$$\log (\bar{M}/\bar{V}) = \log (\bar{M}/\bar{V})$$

$$s_{\log (M/V)} = [\log (M/V)_{\max} - \log (M/V)_{\min}]/2$$

Confidence limits of the true concentration of taxon X/cm³ ($Z = 1.96$ for 0.95 confidence limits; $Z = 2.576$ for 0.99 confidence limits):

$$\log \text{ limit} = [\log \bar{R} + \log (\bar{M}/\bar{V})] \pm Z[(s_{\log R})^2 + (s_{\log (M/V)})^2]^{0.5}$$

Most modern programmable scientific calculators can easily handle these problems. The logarithms can be either common or natural, and the angles can be expressed in degrees or radians. Microcomputers with log and simple trigonometry functions offer extremely fast solutions at relatively low cost. In the event a particular device does not have the arcsin function which is called for in Table I, the angle β can be obtained from the following: $\beta = \arctan \{\sin \beta/[1 - (\sin \beta)^2]^{0.5}\}$.

CORRELATION OF TWO TAXA IN A CORE

Martin and Mosimann (1965) state that the percentages of a taxon A and a taxon B in a sample are correlated because of the constraint that all taxa must sum to 100%. They compute r_c , an estimate of the null model correlation coefficient resulting from the constraint, which for percentages is always negative.

Concentrations of taxa A and B calculated by the spike technique are constrained to have a positive null model correlation coefficient because both concentration estimates are influenced by the value μ/ν happened to take in the sample. The constraint correlation coefficient for the logarithms of the concentrations of taxa A and B in a single sample is $r_c = \delta/[\delta^2 + \omega_A^2 + \omega_B^2]^{0.5}$, which expresses the fact that of the total variance involved in the estimates of the two concentrations, $\delta^2/(\delta^2 + \omega_A^2 + \omega_B^2)$ is the proportion that is common to both.

For the samples from a segment of core, the logarithms of the concentration estimates of the two taxa should be used to calculate a product moment correlation coefficient r . The value for $r_c^2 = \Sigma \delta^2 / (\Sigma \delta^2 + \Sigma \omega_A^2 + \Sigma \omega_B^2)$, the positive square root of which is the constraint coefficient r_c . The procedure in Steel and Torrie (1960, pp. 188–189) is convenient for testing whether an observed r is significant. This involves Fisher's z transformation of the observed r , calculating a 0.95 confidence limit for z , and converting the limits back to r . If the 0.95 limits of r do not include r_c , one concludes that the power function relating the concentrations of taxa A and B differs significantly from that owing to the constraint. The analyst should see that the necessary minimum numbers of grains of taxa A and B are present in all the samples to insure that the $s_{\log R}$'s are reasonable estimators of ω_A and ω_B . A computer can then calculate and test the results without further intervention.

A NULL TEST

Given two samples for which 0.95 confidence intervals for true pollen concentration γ have been calculated, how can one decide whether the samples' concentration estimates $\bar{RM}/\bar{V}$ are too different to have come from the same population? Put another way, how can one test the null hypothesis that the difference between the two estimates results simply from chance variability that would be expected in samples with the same true pollen concentration? How much difference, for example, can be accepted between adjacent samples in the Lake Superior core (Fig.13) before one must conclude a real change in pollen concentration has occurred?

Mosimann (1965, p.662) provides a χ^2 test of the hypothesis that the true ratio u is the same for two or more samples. His test should not be applied to our problem, however, because Mosimann used indigenous pollen taxa as marker grains. The test does not consider the extra variance introduced when the exotic markers are added to the sample; it does not allow for the fact that μ/ν varies between samples.

We need to consolidate some symbols. Let $\log C$ represent the concentration variable $\log (RM/V)$, let $\log \bar{C}$ be $\log (\bar{RM}/\bar{V})$, and let $\sigma^2 = \delta^2 + \omega^2$. Assume two samples for which the pollen concentration estimates differ: $\log C_2 > \log C_1$. If $\log C_2 \sim N[\log \gamma_2, \sigma_2^2]$ and $\log C_1 \sim N[\log \gamma_1, \sigma_1^2]$, then $\log C_2 - \log C_1 \sim N[\log \gamma_2 - \log \gamma_1, \sigma_2^2 + \sigma_1^2]$. Setting $Z = 1.96$, we can make the following probability statement:

$$p[\overline{\log C_2} - \overline{\log C_1} - Z(\sigma_2^2 + \sigma_1^2)^{0.5} \leq \log \gamma_2 - \log \gamma_1 \leq \overline{\log C_2} - \overline{\log C_1} + Z(\sigma_2^2 + \sigma_1^2)^{0.5}] = 0.95.$$

If $\log \gamma_2 = \log \gamma_1$, that is, if the pollen concentration is the same in both samples, then the term on the left will be ≤ 0 . The left term can only have a value > 0 when $\log \gamma_2 > \log \gamma_1$. In that case, of course, $\gamma_2 \neq \gamma_1$, and therein lies a basis for testing the null hypothesis that the true pollen concentration in the two samples is the same.

My confidence interval for a sample's pollen concentration is taken from the distribution of the JPD of $\log C = \log (RM/V)$ which in its logarithmic form is a normal distribution; when coded in Z units (Fig.11), it is a standard normal distribution. This underlying distribution can be recovered from the final, say, 0.95 confidence limits by reconvertng the limits to log values. $\log \bar{C}$ is half the sum of the log limits; $s_{\log C}$, the estimator for σ , is half the difference between the log limits divided by 1.96, the value of Z used in establishing the limits.

To illustrate the problem, let us compare two synthetic samples in the upper part of Fig.12, specifically the 2nd and 9th samples from the left. The 0.95 confidence intervals for these two samples both just contain the defined true pollen concentration in the hypothetical sediment, although the samples' concentration estimates $RM/\bar{V}$ lie on opposite sides of the true value. The limits are shown in Fig.14. The original concentration estimates in these samples will be the geometric means of the limits. Let the confidence interval with the greater concentration estimate (136,633 grains/cm³) be C_2 and that with the smaller estimate (105,845 grains/cm³) be C_1 . By converting their limits to logarithms we can restore the underlying normal distribution upon which the limits are based and determine $\log C_2$, $s_{\log C_2}$, $\log C_1$, and $s_{\log C_1}$. The resultant distributions are coded in standard deviation or Z units for joint probability estimation, and plotted at the left ($\log C_2$) and top ($\log C_1$) sides of Fig.14.

We are interested in the quantity $\log C_2 - \log C_1$ and its distribution across the face of the figure; the diagonal lines show the loci of selected values. The lines are parallel; they represent values of $\log C_2 - \log C_1$ that decrease from upper left to lower right. At some point they become negative. The gradient of $\log C_2 - \log C_1$ extends at right angles to the isolines. Its orientation can be predicted from $\theta = \arctan [s_{\log C_2}/s_{\log C_1}]$. Here, $\theta = \arctan [0.0355/0.0366] = 44.1^\circ$, measured from a line parallel to the C_1 scale and passing through the figure's origin.

Given the two normal populations, we see that 0.95 of the JPD for $\log C_2 - \log C_1$ lies between limits $\pm 1.96 Z$ units of the origin, and 0.99 of the JPD falls between $\pm 2.576 Z$ units of the origin. The placement of these points is shown by dots and small circles on Fig.14.

We can decode the Z unit coordinates of these points to log units by multiplying the Z dimensions by their respective standard deviations and adding their respective means. We are not really interested in the upper or maximum limit of $\log C_2 - \log C_1$:

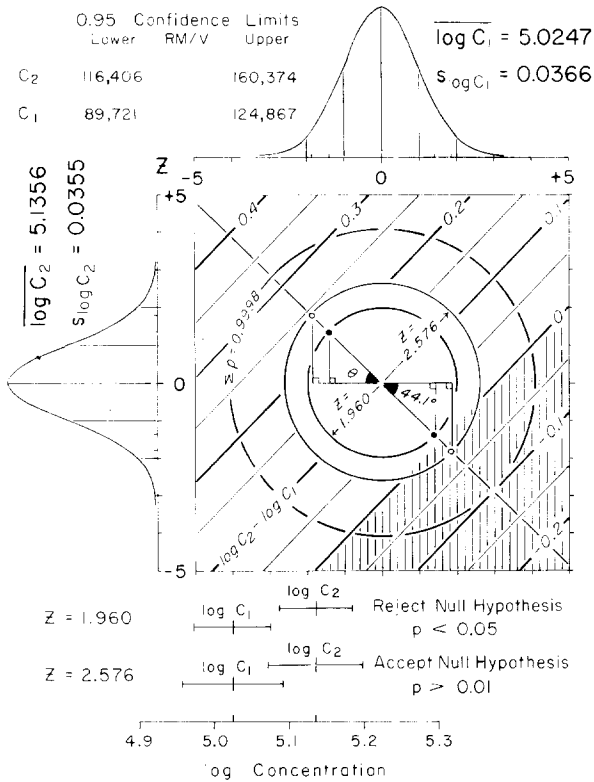


Fig.14. The geometric basis for testing the null hypothesis that the pollen concentration is the same in two samples. The illustration uses the 2nd and 9th synthetic samples at the upper left of Fig.12.

$$(\log C_2 - \log C_1)_{\max} = \overline{\log C_2} - \overline{\log C_1} + Z[(s_{\log C_2})^2 + (s_{\log C_1})^2]^{0.5}$$

which is 0.2109 for $Z = 1.96$ and 0.2423 for $Z = 2.576$. Rather we wish to know the minimum value:

$$(\log C_2 - \log C_1)_{\min} = \overline{\log C_2} - \overline{\log C_1} - Z[(s_{\log C_2})^2 + (s_{\log C_1})^2]^{0.5}$$

In Fig.14 we see that for $Z = 1.96$, $(\log C_2 - \log C_1)_{\min}$ is positive (0.0109) and for $Z = 2.576$, $(\log C_2 - \log C_1)_{\min}$ is negative (-0.0205). Knowing this we can immediately reject, at the $p = 0.05$ level of confidence, the null hypothesis that the two pollen samples could have come from sediment with the same pollen concentration because $p < 0.05$. On the other hand, at the $p = 0.01$ level (here, $p > 0.01$), we can accept the null hypothesis that the two concentration estimates could have come from the same population. At least this is true at the resolution we can achieve, based on the precision of the samples' volumes and spikes and the size of the pollen counts.

To clarify the reasoning, the solid dots and the open circles mark the limits of $\log C_2 - \log C_1$ for, respectively, $Z = 1.96$ and $Z = 2.576$. The points can be resolved into their equivalent values of $\log C_1$ and $\log C_2$. For a specified

Z , the two values for $\log C_1$ lie symmetrically about $\overline{\log C_1}$, and the two values of $\log C_2$ lie symmetrically about $\overline{\log C_2}$. These two pairs of intervals are plotted on the bar graph at the bottom of Fig.14. Note that the difference between the upper limit of $\log C_2$ and the lower limit of $\log C_1 = (\log C_2 - \log C_1)_{\max}$; the difference between the lower limit of $\log C_2$ and the upper limit of $\log C_1 = (\log C_2 - \log C_1)_{\min}$. It can be shown that the two intervals of $\log C_1$ and $\log C_2$ in the upper pair ($Z = 1.96$) contain that portion of each of the individual distributions which together make up 0.95 of their JPD. Similarly, the two intervals in the lower pair ($Z = 2.576$) contain that portion of distributions $\log C_1$ and $\log C_2$ which together make up 0.99 of their JPD.

In the general case, if more than 0.95 [0.99] of the JPD is found where the two populations have no common values, then less than 0.05 [0.01] of the JPD can be common to both populations. We can reject the null hypothesis at the $p = 0.05$ [0.01] level because $p < 0.05$ [0.01].

However, if 0.95 [0.99] or less of the JPD is found where the two populations have no values in common, then 0.05 [0.01] or more of the JPD is common to both populations. We can accept the null hypothesis at the 0.05 [0.01] level because $p \geq 0.05$ [0.01].

The two concentrations, it will be recalled, were the 2nd and 9th synthetic samples of Fig.12. Therefore, although they did indeed come from the same pollen concentration population, we would expect that such an extreme difference in their concentration estimates $\overline{RM}/\overline{V}$ would occur in less than 5% of the sample pairs. On the other hand, we should not be surprised to find such a difference in somewhat more than 1% of the pairs.

The 200 samples of Fig.12 can be combined into $200 \times 199/2 = 19,900$ different pairs. In testing at the $p = 0.05$ level, the null hypothesis could be accepted in 96.0% of the comparisons. Testing at the $p = 0.01$ level, the null hypothesis was accepted in 99.4% of the trials. The null test functions as expected.

Table II summarizes the steps necessary to apply the null test when the samples' concentration confidence limits are known. If the quantities $\overline{\log C}$ and $s_{\log C}$ were saved when the confidence limits were originally calculated, of course these may be used directly.

We can apply the procedure to the Lake Superior data of Fig.13 to test the null hypothesis that the pollen concentration is the same in adjacent levels of the core. The results are shown at the right margin. We can infer that the measured pollen concentration estimates in certain parts of the core, say from 30 to 34 cm, do not differ significantly — at least at the resolution that is possible from the sample measurements. But the core's pollen concentration does seem to change significantly at other places — for example, between 34 and 35 cm. Here the analyst may suggest that the quantity of pollen reaching the sediment has changed, that there has been a change in pollen preservation, that there has been an alteration in the rate of sedimentation, or that something else has happened. It may be impossible to discover the actual reason for the different concentration, but at least the analyst can

TABLE II

A test of the hypothesis that the microfossil concentration is the same in two different samples

-
- Given:
1. Confidence limits for the samples' microfossil concentration.
 2. If 0.95 confidence intervals are involved, $Z' = 1.96$; if 0.99 confidence intervals are involved, $Z' = 2.576$.
 3. Let C_2 represent the confidence interval with the larger mean concentration estimate $\overline{RM}/\overline{V}$, and let C_1 represent the confidence interval with the smaller mean estimate. Define the upper and lower confidence limits of C_2 , respectively, as $(C_2)_{\max}$ and $(C_2)_{\min}$; define the upper and lower confidence limits of C_1 , respectively, as $(C_1)_{\max}$ and $(C_1)_{\min}$.

$$\overline{\log C_2} = [\log (C_2)_{\max} + \log (C_2)_{\min}] / 2$$

$$s_{\log C_2} = [\log (C_2)_{\max} - \log (C_2)_{\min}] / (2Z')$$

$$\overline{\log C_1} = [\log (C_1)_{\max} + \log (C_1)_{\min}] / 2$$

$$s_{\log C_1} = [\log (C_1)_{\max} - \log (C_1)_{\min}] / (2Z')$$

Test: To test whether the null hypothesis can be accepted at $p \geq 0.05$, $Z = 1.960$.

To test whether the null hypothesis can be accepted at $p \geq 0.01$, $Z = 2.576$.

$$(\log C_2 - \log C_1)_{\min} = \overline{\log C_2} - \overline{\log C_1} - Z[(s_{\log C_2})^2 + (s_{\log C_1})^2]^{0.5}$$

Accept null hypothesis if $(\log C_2 - \log C_1)_{\min} \leq 0$.

Reject null hypothesis if $(\log C_2 - \log C_1)_{\min} > 0$.

spend his time studying levels in the core where it is unlikely that such a difference could have occurred simply by chance.

COMBINING SAMPLES

There are many occasions when an analyst needs to combine independent observations into some sort of average and to assign confidence limits to the result. A rather trivial case that comes to mind involves the amalgamation of pollen and marker counts from several slides that were made from a vial of processed material. In this case the several slides only serve to extend our observations over a larger portion of the vial's pollen and marker population. The combined $\hat{u}$ is the sum of all the indigenous grains divided by the sum of all the markers; Mosimann's equation then gives the confidence limits of u .

But a different situation arises when we wish to combine counts from independent sediment samples from a single level. I have said that the prudent analyst should take duplicate samples at some levels in the core and use one member of the pair to get a rough idea of the number of marker grains that should be added for efficient counting. We should be able to profit from that extra labor. In a somewhat different case, we may have a core with several pollen zones and wish to calculate the average concentration of a taxon in each of the zones. In both of these cases we would err if we tried to get an average $\hat{u}$ by summing the indigenous grains and dividing by the sum of the marker grains. We would also err if we tried to establish an average by calculating the mean of the samples' concentration estimates. We will see below

that the best measure of central tendency in these situations is the geometric mean of the concentration estimates. (The geometric mean is the antilog of the arithmetic mean of the logarithms of the concentration estimates.) But more importantly we need a method of establishing confidence limits for the true average pollen concentration in N such samples.

Consider the case where we have two independent samples from the same level in a core. If $\log C_1 \sim N[\log \gamma_1, \sigma_1^2]$ and $\log C_2 \sim N[\log \gamma_2, \sigma_2^2]$, then:

$$\log C_1 + \log C_2 \sim N[\log \gamma_1 + \log \gamma_2, \sigma_1^2 + \sigma_2^2].$$

Setting $Z = 1.96$ we can make the following probability statement in which each term of the inequality is divided by N , the number of independent samples involved:

$$p\{[\overline{\log C_1} + \overline{\log C_2} - Z(\sigma_1^2 + \sigma_2^2)^{0.5}]/N \leq (\log \gamma_1 + \log \gamma_2)/N \leq [\overline{\log C_1} + \overline{\log C_2} + Z(\sigma_1^2 + \sigma_2^2)^{0.5}]/N\} = 0.95.$$

Let us use the 0.95 confidence limits of the two samples shown in Fig.14 (the 2nd and 9th synthetic samples of Fig.12), convert them to logarithms and recover $\log C$ and $s_{\log C}$, the estimator of σ . If their values are substituted into the above probability statement, we can conclude with 0.95 confidence that the interval from 107,200 to 134,900 grains/cm³ includes the level's true pollen concentration. This interval is substantially smaller than the confidence intervals of the individual samples, and it does indeed contain 120,600 grains/cm³, the defined concentration of the artificial sediment.

The reader who remembers that these two synthetic samples were specifically chosen because their concentration estimates were spaced on opposite sides of the true value may feel uncomfortable with this example. In such a case any averaging technique would likely include the true value. It is important to realize, however, that if sample pairs were randomly selected from those in the upper part of Fig.12, and if 0.95 confidence limits were calculated for each pair, very nearly 95% of such intervals would include the true concentration γ . This also applies to the samples in the lower part of Fig.12. And in this example the pairs could include a sample from both populations. I also emphasize that the combined interval's stated probability of including γ applies whether or not the individual samples' confidence intervals actually contain γ .

Although we have been considering confidence limits for combined pairs of samples from a single level, the logic expressed in the above probability statement does not require that the samples all come from sediment with the same concentration γ or that N be restricted to 2. By referring to the probability statement, it will be seen that the log limit for the combined samples will be based on a distribution with mean $[\sum_{i=1}^N \overline{\log C_i}]/N$ and standard deviation $[\sum_{i=1}^N \sigma_i^2]^{0.5}/N$. If 0.99 confidence limits are desired, for example, the standard deviation should be multiplied by $Z = 2.576$. When this quantity is subtracted from the mean, the antilog of the result is the lower confidence limit. When this quantity is added to the mean, the antilog of the sum is the

upper confidence limit. And as was noted earlier, the antilog of the log mean itself will be the geometric mean of the concentration estimates of the N samples that were combined. Table III summarizes the steps used to calculate confidence limits for combined samples.

STRATEGIES TO REDUCE THE CONFIDENCE INTERVAL

We need to consider possible strategies for reducing the 0.95 confidence interval for a sample's pollen concentration γ . Is it more efficacious to decrease the variability of M/V by using multiple marker tablets and sediment aliquots, or should the variability of R be reduced by counting more grains?

The three columns of Fig.15 represent different hypothetical sediments for which we have assumed counts of x and n , hence $\bar{R}$. In the left column the sample $\bar{R} = 0.33$; $\bar{R} = 1$ in the center column, and $\bar{R} = 2$ in the right column. This is not to imply that ρ is known for any of the 12 samples. Rather I assume these unlikely but possible counts so that, for purposes of comparison, the pollen concentration estimates $\bar{R}\bar{M}/\bar{V}$ for the samples in each column will be the same. The marker grain count is 300 in rows A and C, and 1200 in rows B and D. Each of the samples in rows A and B contains one of our sediment aliquots and a single marker tablet. For these six samples $\bar{M} = 12,489$, $s_m = 491$, $\bar{V} = 0.2071$, and $s_v = 0.0037$. Each of the samples in rows C and D contains four sediment aliquots and four marker tablets; $\bar{M} = 49,956$, $s_m = 982$, $\bar{V} = 0.8284$, and $s_v = 0.0074$.

Pollen concentration estimates $\bar{R}\bar{M}/\bar{V}$ in the four samples in the left column are all 20,100 grains/cm³; the concentration estimates for the central and right columns are, respectively, 60,300 and 120,600 grains/cm³. These values

TABLE III

Confidence limits for microfossil concentration in a sample composed of N independent samples

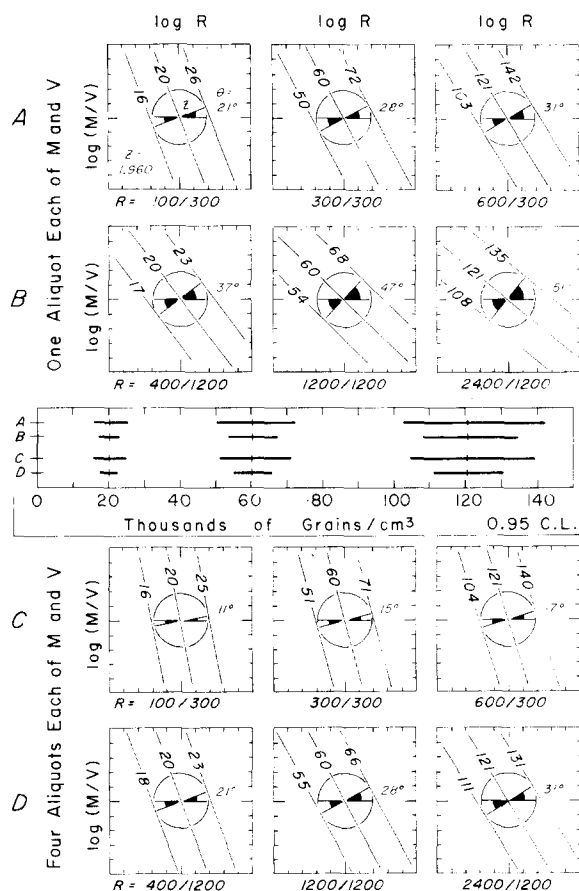
-
- Given:
1. Confidence limits for the true microfossil concentration γ_i in each of N samples, where $i = 1, 2, \dots, N$.
 2. If 0.95 confidence intervals are involved, $Z' = 1.96$; if 0.99 confidence intervals are involved, $Z' = 2.576$.
 3. Let C_i represent the confidence interval for γ_i whose concentration estimate $\bar{C}_i = \bar{R}_i \bar{M}_i / \bar{V}_i$. Define the upper and lower confidence limits of C_i , respectively, as $(C_i)_{\max}$ and $(C_i)_{\min}$.

$$\overline{\log C_i} = [\log (C_i)_{\max} + \log (C_i)_{\min}] / 2$$

$$s_{\log C_i} = [\log (C_i)_{\max} - \log (C_i)_{\min}] / (2Z')$$

Confidence limits of the true concentration of taxon X/cm³ in a sample composed of N independent samples ($Z = 1.96$ for 0.95 confidence limits; $Z = 2.576$ for 0.99 confidence limits):

$$\log \text{ limit} = \left\{ \sum_{i=1}^N (\overline{\log C_i}) \pm Z \left[\sum_{i=1}^N (s_{\log C_i})^2 \right]^{0.5} \right\} / N$$



Should the analyst conclude that no useful purpose is gained by attempting to reduce the variance of $\log (M/V)$? The best option is to keep all variances as small as practicable. The smallest confidence intervals do occur in row D in which the samples combine multiple aliquots and high pollen counts. It requires very little effort to add a few marker tablets to some sediment aliquots. It does require effort to increase the pollen counts. One should keep the counts as large as possible, but not neglect the distribution of $\log (M/V)$.

Fig.15 illustrates angle θ which is a measure of the gradient between the log limits. Although we do not need to know θ to compute the limits algebraically, it provides useful information about the distributions $\log (M/V)$ and $\log R$ that is not apparent from the raw data. Angle θ is a function of the ratio of the two distributions' standard deviations, and it is also a function of the ratio of their variances:

$$\theta = \arctan (s_{\log (M/V)} / s_{\log R}) = \arctan [(s_{\log (M/V)})^2 / s_{\log R}^2]^{0.5}$$

If $\theta = 45^\circ$ the variances of the distributions of $\log (M/V)$ and $\log R$ are equal. When $\theta < 45^\circ$ the variance of $\log (M/V)$ is less than that of $\log R$, and when $\theta > 45^\circ$ the variance of $\log (M/V)$ is greater than $\log R$. The variable with the greater variance will limit how small we can hope to make the confidence interval for concentration.

In Fig.15, $\log R$ is the limiting variable in all but two cases. Whenever $\theta < 45^\circ$ the effective way to reduce the confidence interval is to increase the counts. But note that $\theta > 45^\circ$ in the central and right samples of row B. The size of the counts in these two samples is already so high that $\log (M/V)$ has become the limiting variable. This is a rather unusual example where it will not be very productive simply to increase the pollen count unless we can also reduce the variance of $\log (M/V)$.

We have seen that the variable with the largest variance will limit the reduction that can be made in the confidence interval for concentration; this is analogous to the idea that a chain's strength is limited by its weakest link. Depending on what values the variables R , M , and V happen to be, a sample's pollen concentration is the product of R and M/V . If any variable in a product can assume either the value 0 or ∞ , then the product itself may range between 0 and ∞ , and its confidence interval must reflect this fact. In measuring a sediment's concentration, the analyst can influence R by adjusting the number of markers and the sediment volume. He can also control the variance of $\log (M/V)$ and $\log R$ by the size of the pollen count, by making accurate measurements, and by using multiple tablets and sediment aliquots. We have seen that by exercising care, the analyst can avoid situations where any of the variables R , M , or V could ever be 0 or ∞ .

INFLUX AND THE CONCEPT OF AN INFLUX INDEX

An influx estimate $\overline{SRM}/\overline{V}$ incorporates all of the variance inherent in the distributions of R and (M/V) , as well as that resulting from the uncertainties in our knowledge of S . My training as a geologist makes me suspect that S is

a very different variable from the others we have considered. The rates at which sediment accumulates in a basin vary both through time and in space. Water-laden sediment may slump from the side of a basin and end up in the center, reducing the normal rate at one place and increasing it at the other. A single core from such a setting may be very complicated indeed. The total section's thickness will represent the sum of all the different rates that operated for varying lengths of time, further modified by post-depositional compaction. The analyst takes the core as it is; it is not possible to control the variable S as was the case with R , M , and V .

In the rare situation where the sediments are varved, the fortunate worker can count the laminae in the thickness interval from which the pollen sample was taken and get a good estimate of $\bar{S}$ in cm/yr. However, the sediment best suited to pollen concentration studies usually is fine-grained and homogeneous. In such material $\bar{S}$ has to be inferred from carbon-dating a number of levels in the core.

As an example, consider the hypothetical time-depth curve in Fig.16. Assume that a number of carbon dates can be fitted perfectly to some function in which there is an orderly change of age with depth in the core. The pollen analyst who had such a perfect trend would probably be elated, concluding that the lake had a simple sedimentation history. He might conclude, as a first assumption, that the curve is the expected result of a constant rate of sedimentation, and that the slower apparent rates at the bottom of the sequence came about as the older sediments were compacted by the weight of the younger.

The problem is that an average rate of sedimentation over several meters of core does not really provide firm information about the rate in a sample taken from a centimeter interval within the dated sequence. By way of an analogy, I may time my drive to work and compute an average speed; this does not allow me to infer where along the route I was waiting at a stop sign or where I was exceeding the speed limit.

Now assume that the analyst has unlimited funds, and decides to date smaller and smaller intervals (see inset in Fig.16). Even with the best dating techniques possible, and with staggering luck, sooner or later the standard intervals for decay-counting uncertainties will begin to overlap. When that

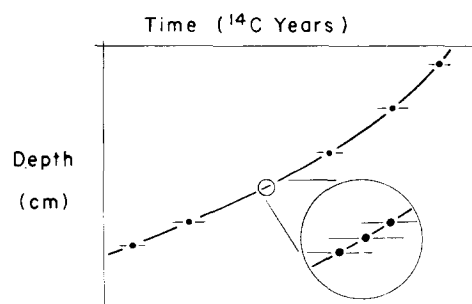


Fig.16. The sedimentation rate variable S is usually inferred from a time—depth curve.

occurs, the analyst has to entertain the possibility that the value of S in a sample may effectively range from near zero to near infinity. S becomes a limiting variable, and the confidence interval for influx approaches infinity. The harder one tries, the worse the results!

I suggest that there is a practical way out of this difficulty based on semantics and common sense. Although pollen influx is a useful concept, a rigorous solution is not possible for a single sample because of uncertainties in the sedimentation rate S . The analyst can still select a best estimate for $\bar{S}$. In Fig.16 this estimate would be the slope of a line tangent to the curve at the depth of the sample. The term "influx index" might be applied to the product of the concentration estimate $\overline{RM}/\bar{V}$ and the sedimentation rate $\bar{S}$. This is exactly what pollen analysts now call influx; the addition of the word "index" should force all parties to recognize that this is not an absolute value. Further, I suggest that the confidence limits for concentration also be multiplied by the assumed constant $\bar{S}$. The limits are a useful way to show, for example, a 0.95 confidence interval for a sample's influx index. By assuming $\bar{S}$ is a known parameter, this confidence interval indicates the inherent uncertainty in the influx index owing to the sample variables R and M/V .

A more satisfying measure can be obtained if the analyst uses multiple samples from a segment of the core; carbon dates can provide a reasonably accurate average sedimentation rate for an interval of sediment. Combine the samples (Table III), computing 0.95 confidence limits for the average pollen concentration in the core segment. Multiply the concentration limits by the geometric mean of the implied sedimentation rates at the depths of the individual samples to obtain a 0.95 confidence interval for the segment's influx index. By working with averages over a longer time span, we reduce the possible variability of S ; this influx index approaches true pollen influx. The same numerical result is obtained from Table III by substituting the individual samples' influx index limits in place of the concentration limits.

CONCLUSIONS

Confidence limits for pollen concentration are useful both to the pollen analyst who wishes to interpret fossil data and to the reader who wants to judge whether the interpretation is reasonable. Statistics cannot help faulty data, but they can show the uncertainties inherent even in data acquired to impeccable standards. The analyst who knows the quality of his measurements is likely neither to ascribe significance to random fluctuations nor to overlook significant trends that do exist. Calculating confidence limits requires information about the samples that is rarely published. The author who respects his audience will include such limits on published diagrams or provide the data needed for their computation.

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REFERENCES

- Beyer, W.H. (Editor), 1968. Handbook of Tables for Probability and Statistics. The Chemical Rubber Co., Cleveland, Ohio, 2nd ed.
- Engstrom, D.R. and Maher, Jr., L.J., 1972. A new technique for volumetric sampling of sediment cores for concentrations of pollen and other microfossils. *Rev. Palaeobot. Palynol.*, 14: 353—357.
- Maher, Jr., L.J., 1972. Absolute pollen diagram of Redrock Lake, Boulder County, Colorado. *Quaternary Res.*, 2: 531—553.
- Maher, Jr., L.J., 1977. Palynological studies in the Western Arm of Lake Superior. *Quaternary Res.*, 7: 14—44.
- Mainland, D., Herrera, L. and Sutcliffe, M.I., 1956. Statistical Tables for Use with Binomial Samples, Contingency Tests, Confidence Limits, and Sample Size Estimates. Department of Medical Statistics, New York University College of Medicine, New York.
- Martin, P.S. and Mosimann, J.E., 1965. Geochronology of Pluvial Lake Cochise, southern Arizona, III. Pollen statistics and Pleistocene metastability. *Am. J. Sci.*, 263: 313—358.
- Mosimann, J.E., 1965. Statistical methods for the pollen analyst. In: B. Kummel and D. Raup (Editors), *Handbook of Paleontological Techniques*. Freeman and Co., San Francisco, Calif., pp.636—673.
- Mosimann, J.E. and Greenstreet, R.L., 1971. Representation-insensitive methods for paleoecological pollen studies. In: G.P. Patil, E.C. Pielou and W.E. Waters (Editors), *Statistical Ecology, 1. Spatial Patterns and Statistical Distributions*. Pennsylvania State University Press, University Park, Penn., pp.23—58.
- Regal, R.R. and Cushing, E.J., 1979. Confidence intervals for absolute pollen counts. *Biometrics*, 35: 557—565.
- Rohlf, F.J. and Sokal, R.R., 1969. *Statistical Tables*. Freeman and Co., San Francisco, Calif.
- Snedecor, G.W. and Cochran, W.G., 1967. *Statistical Methods*. Iowa State University Press, Ames, Iowa, 6th ed.
- Steel, R.G.D. and Torrie, J.H., 1960. *Principles and Procedures of Statistics*. McGraw-Hill, New York, N.Y.
- Stockmarr, J., 1971. Tablets with spores used in absolute pollen analysis. *Pollen Spores*, 13: 615—621.
- Stockmarr, J., 1973. Determination of spore concentration with an electronic particle counter. In: *Geological Survey of Denmark Yearbook 1972*, pp.87—89.